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Research Paper

Optimization of heat sinks for data center server CPUs cooled via single-phase immersion cooling

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ABSTRACT

As modern data centers increasingly support high-performance computing and AI-driven applications, advanced thermal management becomes ever more critical. Single-phase immersion cooling offers an effective and energy-efficient solution for managing high thermal loads. This study focuses on optimizing heat sink geometries and coolant conditions for a dual-CPU server with 24 DIMMs immersed in FC-40 dielectric fluid. Using response surface methodology with a Box-Behnken design (RSM-BBD), this study systematically evaluates the effects of design and operational variables on CPUs temperature, temperature uniformity, and the power required for coolant pumping and thermal restoration. The sensitivity analysis reveals that the coolant flow rate exerts the greatest influence across all optimization criteria. Apart from CPUs temperature, geometric parameters have a more pronounced effect on the remaining optimization metrics than the coolant inlet temperature. The optimized model satisfies all metrics for effective immersion cooling. Specifically, under a thermal design power (TDP) of 400 W, the CPUs temperature is maintained below 60 °C, with the temperature difference between the CPUs kept below 5 °C. Furthermore, the model can handle higher TDPs of up to 600 W while keeping the CPUs temperature below 85 °C. It also performs effectively for servers equipped with advanced chips, such as GaN-based devices, at TDPs up to 800 W. Among all the coolants evaluated, Novec™ 7200 demonstrates the ability to maintain the CPUs temperature at 58.3 °C while requiring the lowest pumping power. Furthermore, the proposed correlations accurately predict the thermal behavior of heat-generating components, with an average absolute deviation of less than 10 %.

1. Introduction

Driven by the rapid advancement of technologies such as artificial intelligence (AI) and high-performance computing (HPC), data centers (DCs) have seen a sharp rise in energy demand, leading to a substantial increase in global electricity consumption [1]. In this context, communication technologies are projected to consume around 20 % of the global electricity by 2030 [2]. At the same time, global DC capacity demand is projected to grow significantly, rising from 60 GW in 2023 to as much as 219 GW by 2030, and potentially reaching 298 GW under high-growth scenarios. Notably, AI-ready DCs, characterized by rapidly increasing power densities, are anticipated to make up approximately 70 % of the total demands by the end of the decade [3]. In parallel with this capacity expansion, the total energy consumption of DCs is expected to increase from 286 TWh in 2016 to 752 TWh by 2030, primarily driven by the substantial power requirements of their cooling systems [4].

DC components consume a substantial amount of electricity, most of which is converted into heat that must be effectively managed to ensure reliable operation [5]. At the same time, ongoing trends in server miniaturization and integration have significantly increased chip power density and operating temperatures. When chip temperatures exceed 75 °C, server failure rates rise exponentially [6]. To mitigate this risk, IT equipment must be kept below critical temperature thresholds, making efficient cooling systems essential for the performance and stability of DCs [7,8]. In recent years, the energy consumption of DCs has been growing at an annual rate of approximately 12 %, accounting for 1.3 %–2 % of global electricity usage [9,10], with cooling systems responsible for 30–50 % of that consumption [11,12]. In this context, optimizing cooling technologies has emerged as a major research focus, not only to

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Nomenclature

c_p	specific heat capacity (J/kg.°C)
g	gravitational acceleration (m^2/s)
h	fin height (mm)
l	fin length (mm)
Q	coolant flow rate (lpm)
p	pressure (Pa)
P_P	pumping power (W)
R	thermal resistance (°C/W)
s	fin spacing (mm)
T	temperature (°C)
t	fin thickness (mm)
u	velocity (m/s)
V	volumetric flow rate (m^3/s)

Greek symbols

μ	dynamic viscosity (Pa.s)
ρ	density (kg/m^3)
λ	thermal conductivity (W/m.°C)

Subscripts

amb	ambient
c	coolant
d	DIMM
in	inlet
out	outlet
m	maximum
s	solid

Acronyms

AAD	Average Absolute Deviation
ANOVA	Analysis of Variance
AI	Artificial Intelligence
BBD	Box-Behnken Design
CCD	Central Composite Design
CPU	Central Processing Units
DC	Data Center
DIMM	Dual Inline Memory Module
GPU	Graphics Processing Unit
HS	Heat Sink
IT	Information Technology
PCB	Printed Circuit Board
PCH	Platform Controller Hub
PSU	Power Supply Unit
RSM	Response Surface Method
SPIC	Single-Phase Immersion Cooling
TDP	Thermal Design Power (W)
TPIC	Two-Phase Immersion Cooling

maintain safe operating temperatures but also to enhance the overall energy efficiency of DCs [13].

Cooling electronic components remains a critical challenge and has been addressed through various strategies, including forced air cooling [14,15], heat pipes [16,17], phase change materials [18,19], and liquid cooling systems [20,21]. Among these, air cooling remains the most widely used method in DCs [8]. Conventional forced air-cooling systems can effectively manage thermal loads up to 37 W/cm^2 [22]; however, they are increasingly inadequate for modern DCs, particularly when compared to advanced liquid cooling methods, that can handle significantly higher chip power densities [10]. In space-constrained 1U servers, advanced heat sink configurations, such as vapor chambers and heat pipes with cut fins, have been shown to reduce thermal resistance and enhance cooling efficiency, particularly at elevated power levels [23]. Moreover, air-cooling systems require substantial volumetric airflow and high fan power, which not only decreases their overall efficiency but also leads to elevated noise levels and operational costs [24]. Similarly, the cooling capacity of heat pipes is generally limited to localized hotspots, reducing their effectiveness in managing the broader thermal load of an entire system [25]. In this context, liquid cooling offers significant advantages over other approaches and has gained considerable attention in recent years [26,27]. It can be implemented either indirectly via cold plates [28] or directly through immersion cooling [29].

Immersion liquid cooling, a technique in which electronic components are submerged in dielectric fluids, has gained increasing interest due to its enhanced heat dissipation performance and ability to accommodate higher computational densities. Zhang et al. [30] conducted a comprehensive review of immersion cooling technologies, highlighting that, due to its superior heat transfer performance, uniform temperature regulation, and low noise levels, immersion cooling has emerged as an effective solution for the thermal management of high heat flux electronic devices. This technology is generally classified into two main categories: single-phase immersion cooling (SPIC) and two-phase immersion cooling (TPIC). While TPIC offers a significantly higher heat transfer coefficient, reportedly improving performance by 72–79 % [31], its widespread adoption is hindered by several challenges, including the high cost of coolants [29], environmental concerns [32], and reliability issues [33]. In contrast, SPIC is recognized for its simplicity, lower cost, and greater environmental compatibility, despite having a relatively lower heat transfer coefficient (<2 kW/m^2.K) [31]. Notably, Wang et al. [34] demonstrated that a pump-driven SPIC system reduced the peak temperature and thermal variability of electronic components by 39.4 % and 74 %, respectively. These advantages underscore the potential of SPIC as a practical and scalable thermal management solution for modern high-performance data centers.

Bansode et al. [35] confirmed that servers fully enclosed in cooling fluids can operate efficiently under elevated temperature conditions. Multiple studies have indicated that SPIC systems improve power usage effectiveness. Supporting these findings, Hnayno et al. [36] evaluated the performance of conventional air cooling versus SPIC in high heat flux servers, showing that SPIC reduces power consumption per server by at least 20 %, thereby demonstrating its substantial energy-saving potential. Additionally, forced convection systems have exhibited superior performance compared to natural circulation systems in SPIC setups [37]. Taddeo et al. [38] employed both experimental and numerical approaches to demonstrate that SPIC provides an effective solution for cooling server tanks containing 54 servers, each rated at 400 W. Due to the risk of electrical hazards, advanced immersion cooling systems generally employ dielectric liquids as the coolant [39].

Among the various dielectric fluids, fluorocarbons, hydrofluoroethers, engineered coolants, and oils are the most commonly used in SPIC systems. Jithin et al. [40] compared the cooling performance of deionized water, mineral oil, and e-fluoride, identifying deionized water as the most effective for controlling electronic component temperatures. Cheng et al. [41], in their experiments using 3 M Novec 7100, investigated the effects of different flow rates and heat sink materials. Their findings revealed that lower flow rates resulted in higher CPU temperatures and less uniform cooling, whereas increased flow rates significantly enhanced thermal performance, emphasizing the critical role of coolant properties and flow dynamics in system optimization. Wen et al. [42] designed five novel nanofluids based on FC-40 and performed simulations to assess their performance in a SPIC system. Among them, Al-FC40 exhibited the best thermal performance, reducing peak CPU

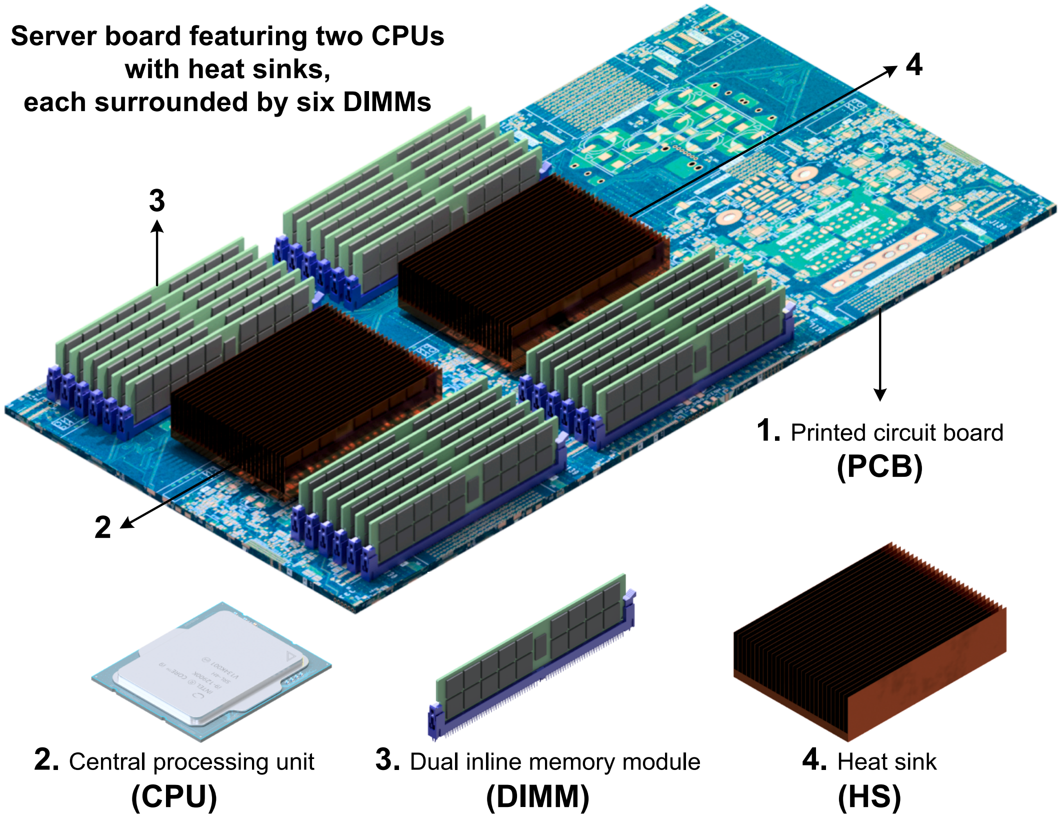

Fig. 1. An illustrative schematic of 1U rack server board.

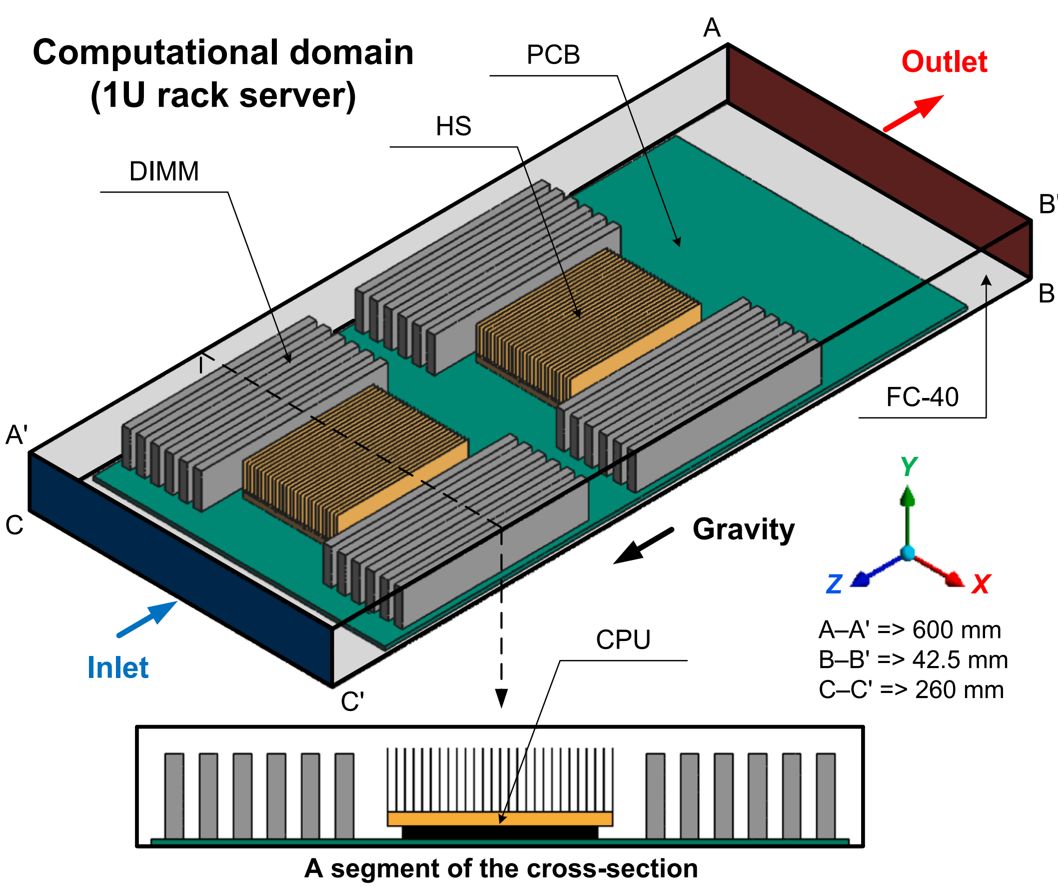

Fig. 2. Computational domain for the 1U rack server configuration employed in the single-phase immersion cooling study.

Table 1
Technical specifications of the components in the 1U rack server.

Component	Printed circuit board (PCB)	Central processing unit (CPU)	Dual inline memory module (DIMM)	Heat sink (HS)
Function	Mechanical support and electrical connections	Performing calculations and severing as a major heat source	Providing volatile memory for temporary data storage and serving as a minor heat source	Absorbing and dissipating heat from CPUs
Number	1	2	24	2
Dimension [mm]	250 × 3 × 500	70 × 4.5 × 75	6.5 × 30 × 160	80 × 22.5 × 112.5
Power [W]	0	100–800	15	0
Material	FR-40	Silicon	Silicon	Copper
Density [kg/m^3]	1850	2300	2300	8978
Thermal conductivity [W/m·s]	0.3	98.4	98.4	387.6
Specific heat [J/kg·K]	1000	721	721	381

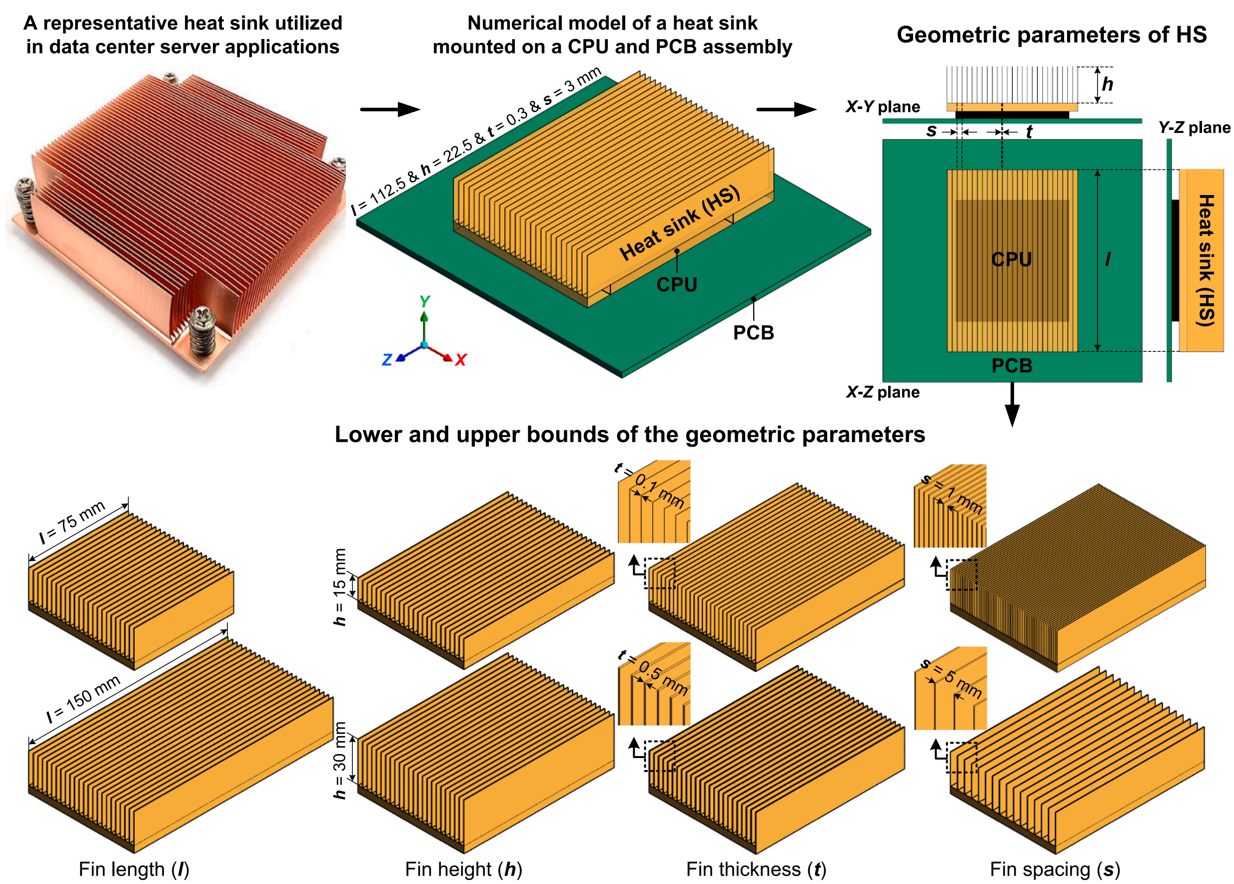

Fig. 3. Geometric structures of the straight-fin heat sink.

Table 2
Optimization factors along with their symbols, units, and corresponding levels.

Factors	Symbols	Units	Levels		
			−1	0	+1
A: Fin length	l	mm	75	112.5	150
B: Fin height	h	mm	15	22.5	30
C: Fin thickness	t	mm	0.1	0.3	0.5
D: Fin spacing	s	mm	1	3	5
E: Coolant flow rate	Q	lpm	2	5	8
F: Coolant inlet temperature	T_{in}	°C	20	30	40

temperature by 6.5 % owing to its superior thermal conductivity. Despite these advancements, experimental studies in this field remain limited, primarily focusing on operating conditions, coolant selection, and heat dissipation enhancement. Shrigondekar et al. [43] systematically analyzed the thermal performance of SPIC in a 1U server using dielectric fluids (PAO-6 and FC-40), highlighting the significant influence of system parameters such as inlet/outlet configuration, fin pitch, fluid bypass, and viscosity-driven flow characteristics on heat transfer efficiency.

While several studies have investigated the performance of SPIC systems, the current body of literature remains limited, particularly in addressing the escalating thermal demands of modern data center hardware driven by rapid advances in AI and HPC. Most existing studies have primarily focused on low- to medium-TDP (Thermal Design Power) processors, with insufficient attention paid to high-TDP scenarios, which pose more severe thermal management challenges [27]. Moreover, due to the high cost and complexity of experimental setups, numerical simulations have emerged as a viable alternative for early-stage design and optimization of SPIC systems [44]. However, conventional heat sink designs, originally developed for air-cooled or indirect liquid cooling applications, have not been systematically re-evaluated for immersion environments, where the thermophysical behavior of dielectric fluids imposes unique design constraints [45]. In particular, the role of fin

Table 3
BBD matrix and corresponding results obtained from numerical simulations.

Test No.	Factors						Responses			
	A: l [mm]	B: h [mm]	C: t [mm]	D: s [mm]	E: Q [lpm]	F: T_{in} [°C]	AverageCPUstemperature [°C]	Maximum CPUs temperatureDifference [°C]	Coolant pumping power $\times 10^{-5}$ [W]	Coolant temperature difference [°C]
1	75	15	0.3	1	5	30	92.49	3.12	19.91	7.12
2	150	15	0.3	1	5	30	85.7	7.15	23.68	7.12
3	75	30	0.3	1	5	30	102.6	7.6	35.29	7.12
4	150	30	0.3	1	5	30	107.68	15.22	40.38	7.12
5	75	15	0.3	5	5	30	81.75	6.27	17.44	7.12
6	150	15	0.3	5	5	30	76.7	11.01	20.9	7.12
7	75	30	0.3	5	5	30	73.57	7.55	26.03	7.12
8	150	30	0.3	5	5	30	66.32	10.53	32.88	7.12
9	112.5	15	0.1	3	2	30	119.78	15.61	2.59	17.84
10	112.5	30	0.1	3	2	30	103.52	24.01	4.51	17.84
11	112.5	15	0.5	3	2	30	107.03	14.01	2.65	17.84
12	112.5	30	0.5	3	2	30	94.8	24.52	4.6	17.84
13	112.5	15	0.1	3	8	30	75.56	4.83	63.46	4.45
14	112.5	30	0.1	3	8	30	68.01	8.28	103.78	4.37
15	112.5	15	0.5	3	8	30	66.43	4.39	64.33	4.45
16	112.5	30	0.5	3	8	30	56.92	8.84	104.01	4.45
17	112.5	22.5	0.1	1	5	20	80.37	8.88	32.9	7.12
18	112.5	22.5	0.5	1	5	20	73.84	8.81	32.97	7.12
19	112.5	22.5	0.1	5	5	20	70.15	10.53	27.38	7.12
20	112.5	22.5	0.5	5	5	20	57.61	9.01	25.87	7.12
21	112.5	22.5	0.1	1	5	40	100.39	8.85	32.87	7.12
22	112.5	22.5	0.5	1	5	40	93.84	8.81	32.97	7.12
23	112.5	22.5	0.1	5	5	40	90.15	10.53	27.38	7.12
24	112.5	22.5	0.5	5	5	40	77.61	9.01	25.87	7.12
25	75	22.5	0.3	1	2	30	124.74	13.52	3.63	17.84
26	150	22.5	0.3	1	2	30	113.47	20.78	4.49	17.84
27	75	22.5	0.3	5	2	30	94.41	15.84	2.95	17.84
28	150	22.5	0.3	5	2	30	89.53	21.61	3.83	17.84
29	75	22.5	0.3	1	8	30	72.68	4.38	87.27	4.45
30	150	22.5	0.3	1	8	30	73.83	8.24	101.78	4.45
31	75	22.5	0.3	5	8	30	70.74	5.1	68.09	4.45
32	150	22.5	0.3	5	8	30	63.31	7.62	84.38	4.45
33	112.5	15	0.3	3	2	20	100.06	14.49	2.58	17.84
34	112.5	30	0.3	3	2	20	86.53	23.76	4.41	17.84
35	112.5	15	0.3	3	8	20	58.45	4.57	62.98	4.45
36	112.5	30	0.3	3	8	20	49.56	8.62	100.1	4.45
37	112.5	15	0.3	3	2	40	120.06	14.49	2.58	17.84
38	112.5	30	0.3	3	2	40	106.53	23.76	4.41	17.84
39	112.5	15	0.3	3	8	40	78.45	4.57	62.98	4.45
40	112.5	30	0.3	3	8	40	69.56	8.62	100.1	4.45
41	75	22.5	0.1	3	5	20	72.63	7.42	26.66	7.12
42	150	22.5	0.1	3	5	20	69.51	13.29	32.61	7.12
43	75	22.5	0.5	3	5	20	57.92	7.4	27.3	7.12
44	150	22.5	0.5	3	5	20	58.95	12.25	33.4	7.12
45	75	22.5	0.1	3	5	40	92.63	7.42	26.67	7.12
46	150	22.5	0.1	3	5	40	89.51	13.29	32.61	7.12
47	75	22.5	0.5	3	5	40	77.93	7.39	27.29	7.12
48	150	22.5	0.5	3	5	40	78.95	12.25	33.4	7.12
49	112.5	22.5	0.3	3	5	30	78.65	10.19	30.15	7.12
50	112.5	22.5	0.3	3	5	30	78.65	10.19	30.15	7.12
51	112.5	22.5	0.3	3	5	30	78.65	10.19	30.15	7.12
52	112.5	22.5	0.3	3	5	30	78.65	10.19	30.15	7.12
53	112.5	22.5	0.3	3	5	30	78.65	10.19	30.15	7.12
54	112.5	22.5	0.3	3	5	30	78.65	10.19	30.15	7.12

geometry in immersion cooling has received limited attention, despite its critical impact on heat transfer efficiency.

The primary novelty of this study lies in addressing this significant gap through a comprehensive and systematic numerical investigation focused on the geometric and operational optimization of heat sinks specifically tailored for SPIC applications. Key geometric parameters, including fin length, fin height, fin thickness, and fin spacing, are evaluated alongside operational variables such as coolant flow rate and inlet temperature, using FC-40 as the working fluid. The performance is assessed based on four key metrics: average CPUs temperature, maximum CPUs temperature difference, coolant temperature rise, and pumping power. To ensure practical relevance, the optimized design is further tested across a wide range of TDP values (200 to 800 W) to demonstrate scalability. Additionally, the model's adaptability is validated with multiple dielectric fluids. Finally, empirical correlations are proposed for predicting the thermal behavior of major heat-generating components (CPUs and DIMMs), offering valuable guidelines for the design and thermal management of SPIC systems in next-generation data centers.

2. Research implementation

2.1. Rack server structure and model configuration

A 1U rack server consists of multiple components, each generating varying amounts of heat and collectively influencing the overall thermal

Table 4
Typical specifications and thermophysical properties of FC-40 within the operating temperature range of 0–100 °C [56].

Typical specifications	Unit		
Appearance	–	Clear and colorless	
Average molecular weight	g/mol	650	
Boiling point at 1 atm	°C	165	
Pour point	°C	−57	
Vaper pressure	Pa	287	
Latent heat of vaporization	J/g	69	
Coefficient of expansion	1/°C	0.0012	
Surface tension	Dynes/cm	16	
Ozone depletion potential	–	0	
Refractive index	–	1.29	
Thermophysical properties	Unit	Correlation	Deviation
Thermal conductivity	W/m^2.°C	$\lambda = 6.5458 \times 10^{-2}$–$(6.3265 \times 10^{-5}T)$+$(1.7444 \times 10^{-8}T^2)$	0.61 %
Specific heat capacity	J/kg.°C	$c_p = 1013.9205$+$(1.9935\ T)$–$(3.1064 \times 10^{-3}T^2)$	0.35 %
Density	Kg/m^3	$\rho = 1897.4313$–$(2.1733\ T)$+$(2.5098 \times 10^{-4}T^2)$	0.11 %
Dynamic viscidity	Pa.s	$\mu = 7.4417 \times 10^{-3}$–$(2.0621 \times 10^{-4}T)$+$(2.9595 \times 10^{-6}T^2)$–$(2.2672 \times 10^{-8}T^3)$+$(7.1023 \times 10^{-11}T^4)$	0.85 %

characteristics of the system [46]. Among these components, CPUs and GPUs are the primary sources of heat generation, particularly in servers designed for AI or HPC applications. This is largely because these units account for a significant portion of the overall power consumption in such systems. Consequently, the thermal management of CPUs and GPUs poses one of the most critical challenges in DS server cooling. In GPU-free servers, CPUs alone can consume over half of the total power, highlighting the urgent need for effective cooling [47]. Accordingly, a substantial body of research has focused on CPU thermal management, as existing literature indicates that the thermal influence of other components on immersion cooling performance is comparatively minor [44]. Additionally, survey data indicate that the majority of DC servers (over 62.6 %) employ a dual-processor configuration [48]. In alignment with these findings, the present study focuses on the thermal management of processors in a rack server equipped with two CPUs.

The physical model employed in this study is based on a third-generation server configuration, featuring two Intel Xeon microprocessors [49]. Each CPU is surrounded by two dual inline memory module (DIMM) blocks, with each block containing six DIMMs, as illustrated in Fig. 1. The model configuration, developed using ANSYS DesignModeler, is illustrated in Fig. 2, while its technical specifications are provided in Table 1. The computational domain dimensions are 600 mm (length, z-axis) × 42.5 mm (height, y-axis) × 260 mm (width, x-axis). A similar server model was previously investigated by Chhetri et al. [50]; however, in their study, the DIMMs were simplified as a single block and assumed to be non-heat-generating components.

2.2. Heat sink designs and coolant characteristics

In the physical model considered in this study, heat sinks are required only for the CPUs, as they generate high power and have a comparatively limited surface area for heat dissipation. The heat sink design under investigation incorporates straight fins, with key geometric parameters optimized to enhance the thermal management of DC servers utilizing SPIC. The optimization process targets four primary geometric parameters: fin length (l), fin height (h), fin thickness (t), and fin spacing (s), as illustrated in Fig. 3. In addition to these design parameters, two operating conditions are also considered: coolant flow rate (Q) and the coolant inlet temperature (T_{in}). These factors are incorporated into the analysis to achieve a comprehensive and effective thermal management strategy.

Previous experimental investigations [51] and numerical studies [52] have shown that matching the height of the heat sink fins to the cabinet height, thereby eliminating the gap between the server and the cabinet's upper wall, enhances coolant flow through the heat sink channels and prevents bypass effects. This improvement reduces thermal resistance, resulting in more effective cooling of the components. However, to explore the interaction between fin height and other design parameters, this study treats fin height as a variable geometric factor.

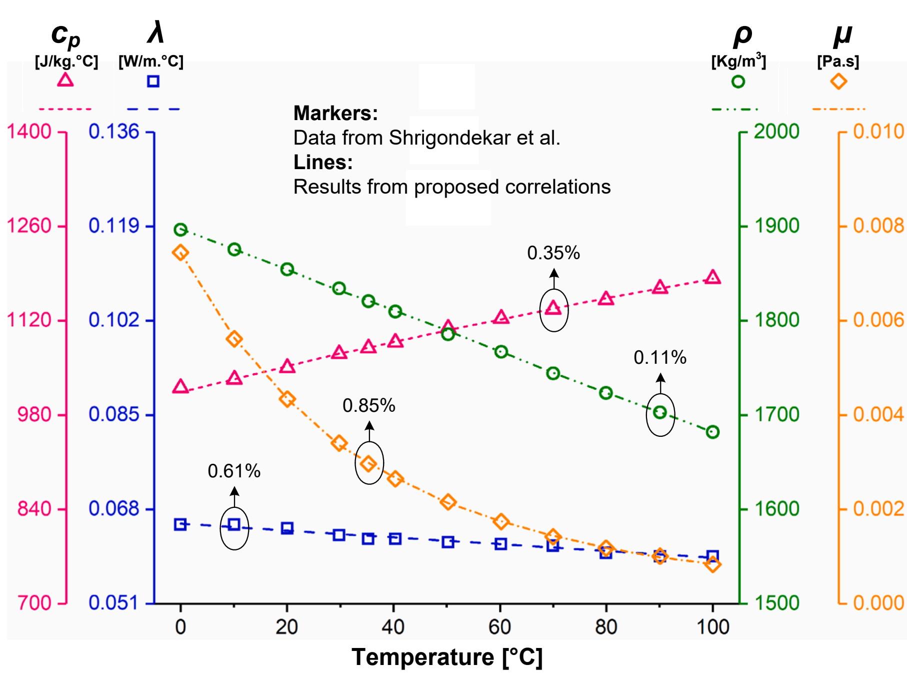

Fig. 4. Accuracy of FC-40 thermophysical property predictions using the proposed correlations.

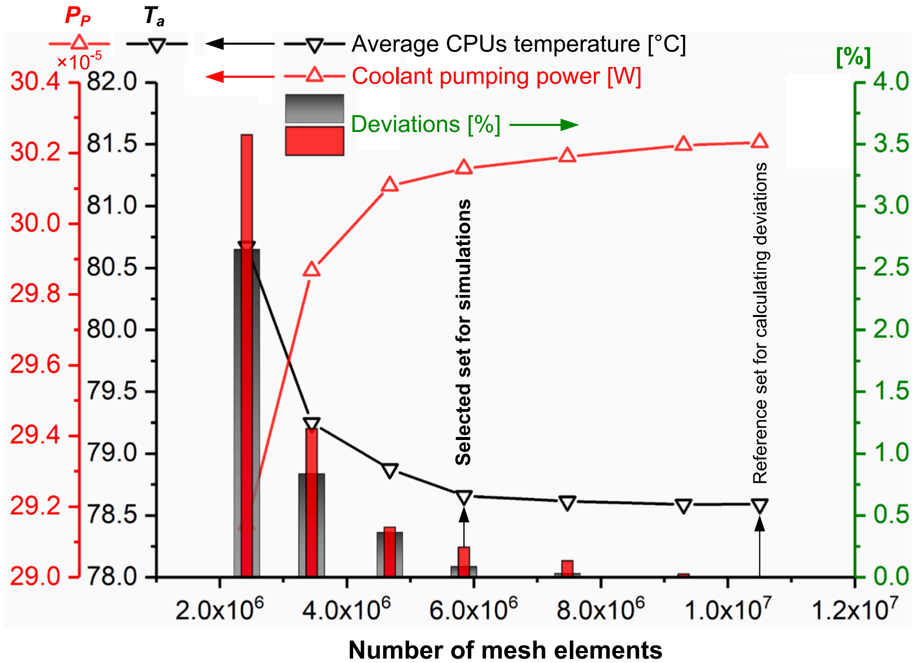

Fig. 5. Variation of average CPUs temperature and coolant pressure drop with the number of mesh elements.

To optimize these factors, the response surface methodology (RSM) is employed. Among the various RSM approaches, Central Composite Design (CCD) and Box-Behnken Design (BBD) are among the most commonly used methods [53]. In this study, as shown in Table 2, all geometric parameters and operating conditions are treated as independent variables to enable a comprehensive evaluation. As a result, a large number of simulations would typically be required. However, the BBD method offers a notable advantage by substantially reducing computational demands, as it requires fewer simulations while still accurately assessing the effects of independent variables on the responses and revealing their interactions [54]. Its reliability and effectiveness have been demonstrated in previous research, with extensive documentation available in the literature [55]. In this study, a total of 54 simulations are performed to analyze key response variables such as the average CPU temperature and coolant pressure drop, as outlined in Table 3.

A quadratic polynomial model is formulated using RSM to characterize how each response variable is influenced by the selected independent variables:

$$Y = \beta_0 + \sum_{i=1}^{k} \beta_i X_i + \sum_{i=1}^{k} \beta_{ii} X_i^2 + \sum_{i}^{k} \sum_{j}^{k} \beta_{ij} X_i X_j + \varepsilon \tag{1}$$

where Y denotes the response, β_0 is the intercept or model constant, β_i are the linear coefficients, β_{ii} are the quadratic coefficients, and β_{ij} represent the interaction coefficients. The terms X_i and X_j are the independent variables, and ε accounts for the random error.

In SPIC systems, dielectric fluids with high boiling points are commonly used to maintain the coolant temperature below the evaporation threshold throughout the cooling process. In this study, FC-40, a thermally stable, fully fluorinated liquid with a boiling point of 165 °C, is employed [56]. Despite its relatively low thermal conductivity, this fluid enhances thermal convection due to its lower viscosity compared to commonly used alternatives [57]. The superior performance of this fluorinated liquid, relative to dielectric oils such as PAO-6, has been demonstrated by Shrigondekar et al. [43]. To estimate the thermophysical properties of FC-40, the correlations presented in Table 4 are developed based on their experimental data. These correlations are valid within the temperature range of 0–100 °C, which fully encompasses the operational temperature range of the coolant in this study. As illustrated in Fig. 4, the proposed correlations demonstrate improved predictive accuracy for the measured values compared to those developed by Wang et al. [58] using linear regression analysis.

2.3. Governing equations and boundary conditions

To streamline the simulation process while ensuring a realistic representation of actual operating conditions, the following assumptions are adopted in the development of the present SPIC model:

- The 3D model focuses primarily on CPUs cooling; therefore, only adjacent DIMMs are included. Components such as the platform controller hub (PCH) and power supply unit (PSU) are omitted due to their downstream placement, low TDP, and greater distance from the CPUs, which according to existing literature [59], result in minimal impact on heat transfer.
- The CPUs and DIMMs are modeled as uniform heat sources.
- The solid components are assumed to have constant physical properties throughout the analysis.
- FC-40 is modeled as a Newtonian fluid.
- The effect of gravity acts in the direction opposite to the coolant flow.
- Due to the inherently complex and turbulent nature of the internal flow, it is modeled as turbulence [24].
- The Boussinesq approximation is used to link Reynolds stresses with mean velocity gradients by introducing an effective turbulent viscosity [37].
- Due to the low Brinkman number, viscous dissipation is considered negligible [52].
- Heat loss to the environment via free convection and radiation is neglected.

Based on these assumptions, the Reynolds averaged Navier-Stokes and energy equations used to model FC-40 flow and heat transfer in the server cooling process via single-phase immersion are expressed as follows [60]:

o Continuity equation:

$$\nabla.(\rho \vec{u}) = 0 \tag{2}$$

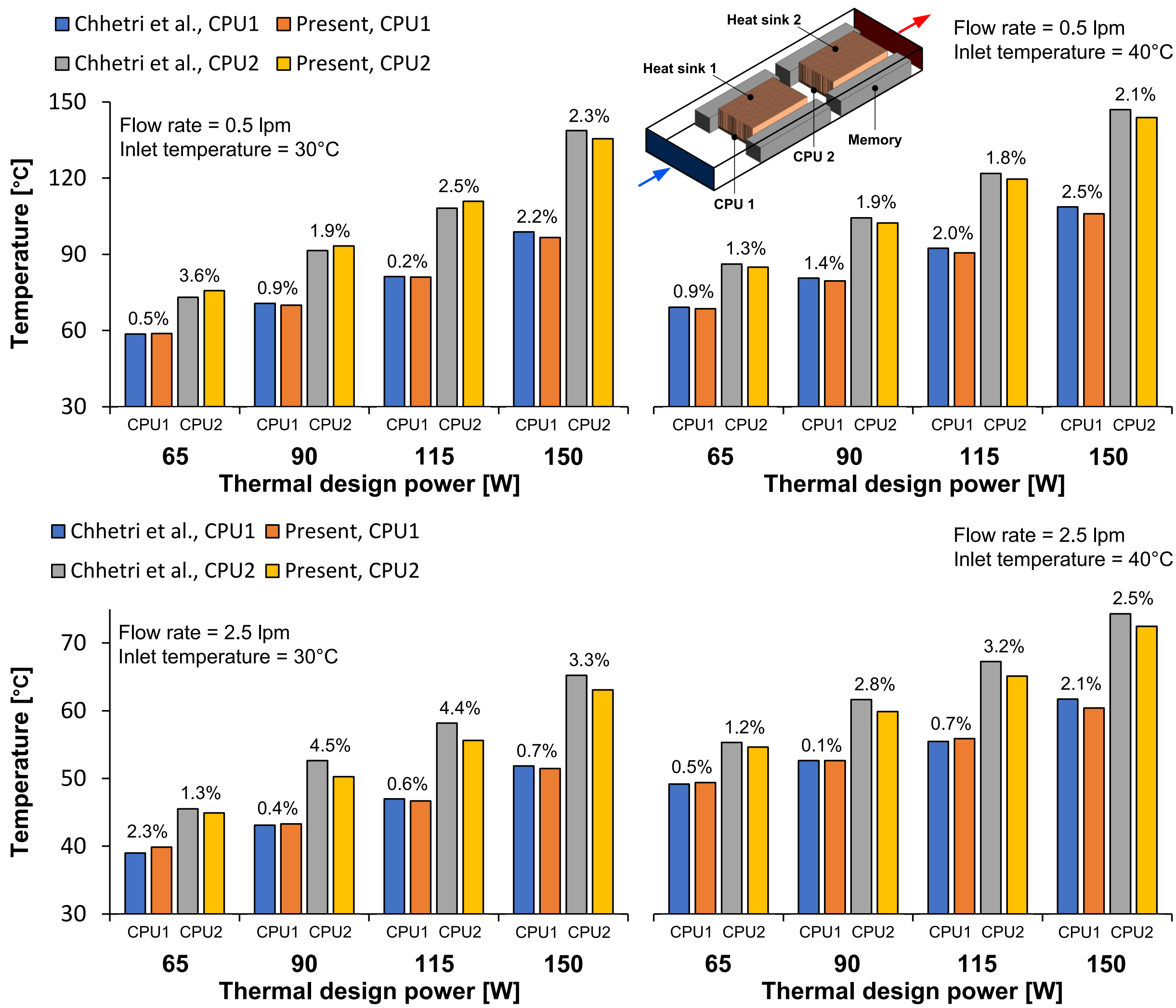

Fig. 6. Physical model used for validation and comparison of results.

o Momentum equations:

$$\nabla.(\rho \vec{u}\vec{u}) = -\nabla p + (\mu + \mu_t)\nabla^2.\vec{u} \tag{3}$$

o Energy equation (for the coolant domain):

$$\nabla.(\lambda \nabla T) = \nabla.(\rho c_p \vec{u}.T) \tag{4}$$

where u, p, and T represent the coolant velocity, pressure, and temperature, respectively, whileμ_t denotes the turbulent viscosity and g signifies gravitational acceleration.

o Energy equation (for the solid domain):

$$\nabla.(\lambda_s \nabla T_s) = 0 \tag{5}$$

where the subscript s refers to the solid domain, i.e., the PCB, CPUs, DIMMs, and HSs.

Compared to the standard k-ε model, the RNG turbulence model provides enhanced accuracy by considering eddy effects and employing analytical formulations for the turbulent Prandtl number and viscosity, particularly in low Reynolds number flows. Although the Realizable model also delivers high accuracy, its greater computational cost and time requirements can limit its applicability. Therefore, the RNG k-ε model is selected in this study to simulate turbulent flow, as it provides an optimal balance between accuracy and computational efficiency [61]. The corresponding equations are presented below:

o k-equation:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k u_i)}{\partial x_i} = \frac{\partial}{\partial x_j}\left(\alpha_k \mu_{eff} \frac{\partial k}{\partial x_j}\right) + G_k + G_b + \rho\varepsilon \tag{6}$$

where α_k is the turbulent Prandtl number, μ_{eff} indicates the effective kinetic viscosity coefficient, and G_k and G_b represent the source terms for turbulent kinetic energy.

o ε-equation:

$$\frac{\partial(\rho\varepsilon)}{\partial t} + \frac{\partial(\rho\varepsilon u_i)}{\partial x_i} = \frac{\partial}{\partial x_j}\left(\alpha_\varepsilon \mu_{eff} \frac{\partial \varepsilon}{\partial x_j}\right) + C_{1\varepsilon}\frac{\varepsilon}{k}(G_k + C_{3\varepsilon}G_b) - C_{2\varepsilon}\rho\frac{\varepsilon^2}{k} - R_\varepsilon \tag{7}$$

where α_ε is the turbulent Prandtl number; $C_{1\varepsilon}$, $C_{2\varepsilon}$, and $C_{3\varepsilon}$ are model constants, and the correction factor (denoted as R_ε) is defined as follows:

$$R_\varepsilon = \frac{C_\mu \rho \eta^3 (1 - \eta/\eta_0)}{1 + \beta\eta^3}\frac{\varepsilon^3}{k} \tag{8}$$

where η is the expansion parameter (the ratio of the turbulent to mean

Table 5
Description of the evaluation metrics.

Index	Average CPUs temperature (T_a)	Maximum CPUs temperature difference (ΔT_m)	Coolant pumping power (P_P)	Coolant temperature difference (ΔT_c)
Significance	A critical indicator of thermal performance, directly influencing system reliability, operational stability, and CPU lifespan	An essential metric for assessing the uniformity of heat distribution across CPUs	A key parameter influencing the total energy consumption and overall thermal management system efficiency	A vital factor for representing the thermal energy required to restore the coolant to its initial state
Feature	Represents the arithmetic mean of the CPUs temperatures	Quantifies the largest observed temperature disparity between CPUs	Represents the mechanical power required to circulate the coolant through the cooling loop	Defines the temperature difference between the coolant's outlet and inlet
Formula	$T_a = (T_{\text{CPU1}} + T_{\text{CPU2}})/2$	$\Delta T_m = (T_{\max} + T_{min})/2$	$P_P = V \times \Delta p$	$\Delta T_c = T_{\text{out}} – T_{in}$
Objective	Maintaining T_a below 60 °C to ensure efficient heat dissipation and avoid performance degradation or thermal stress	Minimizing ΔT_m (below 5 °C) to promote uniform thermal conditions and enhance system reliability	Minimizing P_P (less than 5 % of total system power) to reduce energy consumption	Minimizing ΔT_c (preferably less than 5 °C) to reduce coolant restoration energy

strain time scales), and C_μ, η_0, and β are model constants.

The inlet boundary of the immersion cooling chamber is defined by a specified inlet velocity at a constant temperature, while the outlet boundary is set as a pressure outlet. The CPU and DIMM components are modeled as volumetric heat sources. The liquid–solid interface is treated using a coupled boundary condition, and non-slip conditions are imposed on all solid surfaces. Furthermore, the external surfaces of the server frame are considered adiabatic.

2.4. Solution approach

The governing equations, along with the assigned boundary conditions, are solved iteratively using a pressure-based, double-precision solver within the ANSYS Fluent package (version 2021R2), employing the finite volume method. To handle the pressure–velocity coupling, the pressure correction method based on the SIMPLE algorithm is employed. For the spatial discretization of the computational domain, the pressure term is discretized using a second-order scheme, while the momentum, energy, turbulent kinetic energy, and turbulent dissipation rate equations are discretized using the second-order upwind method. To ensure numerical stability and facilitate convergence, relaxation factors are set as follows: 0.3 for pressure, 1.0 for density, 0.7 for momentum, 0.8 for turbulent kinetic energy, 0.8 for turbulent dissipation rate, 1.0 for turbulent viscosity, and 1.0 for energy. In addition, the iterative solution is initially tested with a residual threshold of 10^{-4} for all variables and gradually refined down to 10^{-8}. The results indicated that reducing the residual below 10^{-6} led to negligible changes (less than 0.02 %) in key output parameters such as the average CPUs temperature and coolant pumping power. Therefore, to balance computational efficiency with solution accuracy, a residual value of 10^{-6} is adopted for all variables.

3. Grid independence analysis and model validation

In this study, ANSYS Meshing is employed to generate the computational grid. The typical mesh configuration consists of structured hexahedral cells in the solid domain and unstructured tetrahedral cells in the fluid domain. To accelerate convergence, reduce computational cost, and maintain solution accuracy, the fluid domain mesh is subsequently converted into a polyhedral mesh using ANSYS Fluent's polyhedral meshing strategy. To assess the impact of grid density on simulation outcomes and ensure grid independence of the results, several grid configurations with varying numbers of elements are analyzed. This investigation is conducted using a model with mid-level geometric parameters ($l = 112.5$ mm, $h = 22.5$ mm, $t = 0.3$ mm, and $s = 3$ mm) and operating conditions ($Q = 5$ lpm and $T_{in} = 30$ °C). As shown in Fig. 5, when the total number of elements exceeds 5.843 million, variations in both the average CPUs temperature and coolant pumping power become negligible. Specifically, increasing the mesh density from 5.843 million to 7.474 million elements results in changes of less than 0.06 % (equivalent to 0.21 °C) in the average CPU temperature and less than 0.11 % in the coolant pumping power. Even when the mesh density is nearly doubled to 10.5 million elements, the variations in these parameters remain minimal, specifically, 0.08 % for the average CPU temperature and 0.24 % for coolant pumping power. Based on these observations, all subsequent simulations are conducted using mesh densities within this validated range to ensure the reliability and accuracy of the numerical results.

To validate the present model and the adopted numerical method, the study conducted by Chhetri et al. [50], which has been extensively used by other researchers [62,63] for validation purposes, was employed as a reference. As depicted in Fig. 6, their investigation focused on SPIC of a server equipped with two CPUs, each operating under different TDP levels (65, 90, 115, and 150 W). Cooling was conducted using EC-110, with inlet temperatures ranging from 25 to 40 °C and flow rates varying between 0.5 and 2.5 lpm. A comparative analysis was performed considering the minimum and maximum coolant flow rates at two different inlet temperatures for each TDP level. The results are summarized in Fig. 6. The findings indicate that, under the specified comparison conditions, the deviation between CPU temperatures predicted by the present model and those reported in the reference study ranges from 0.1 % to 4.5 %. These discrepancies can be attributed to the assumptions made and the differing approaches to computational domain discretization employed in each study. Nevertheless, the relatively low deviation values confirm that the present model is reliable and can be confidently used to predict heat transfer behavior in SPIC systems.

4. Results and discussion

To assess the impact of geometric parameters (l, h, t, and s) and operating conditions (Q and T_{in}) on the thermal management of DC servers utilizing SPIC, four key evaluation indices are considered: (1) the average temperature of CPUs (T_a), (2) the maximum temperature difference among CPUs (ΔT_m), (3) the power consumption required for coolant pumping (P_P), and (4) the coolant temperature difference between outlet and inlet (ΔT_c). A detailed explanation of these evaluation indices is provided in Table 5. Based on these criteria, the following sections present a comprehensive analysis of the results. The effects of design and operational variables on thermal and hydraulic responses are examined, and the optimal configuration for simultaneously achieving the desired performance objectives is identified.

4.1. Optimization results

An analysis of variance (ANOVA) is conducted for the response variables: T_a, ΔT_m, P_P, and ΔT_c. The results are provided in Appendix A and organized as Tables A-1 through A-4. These tables assess the

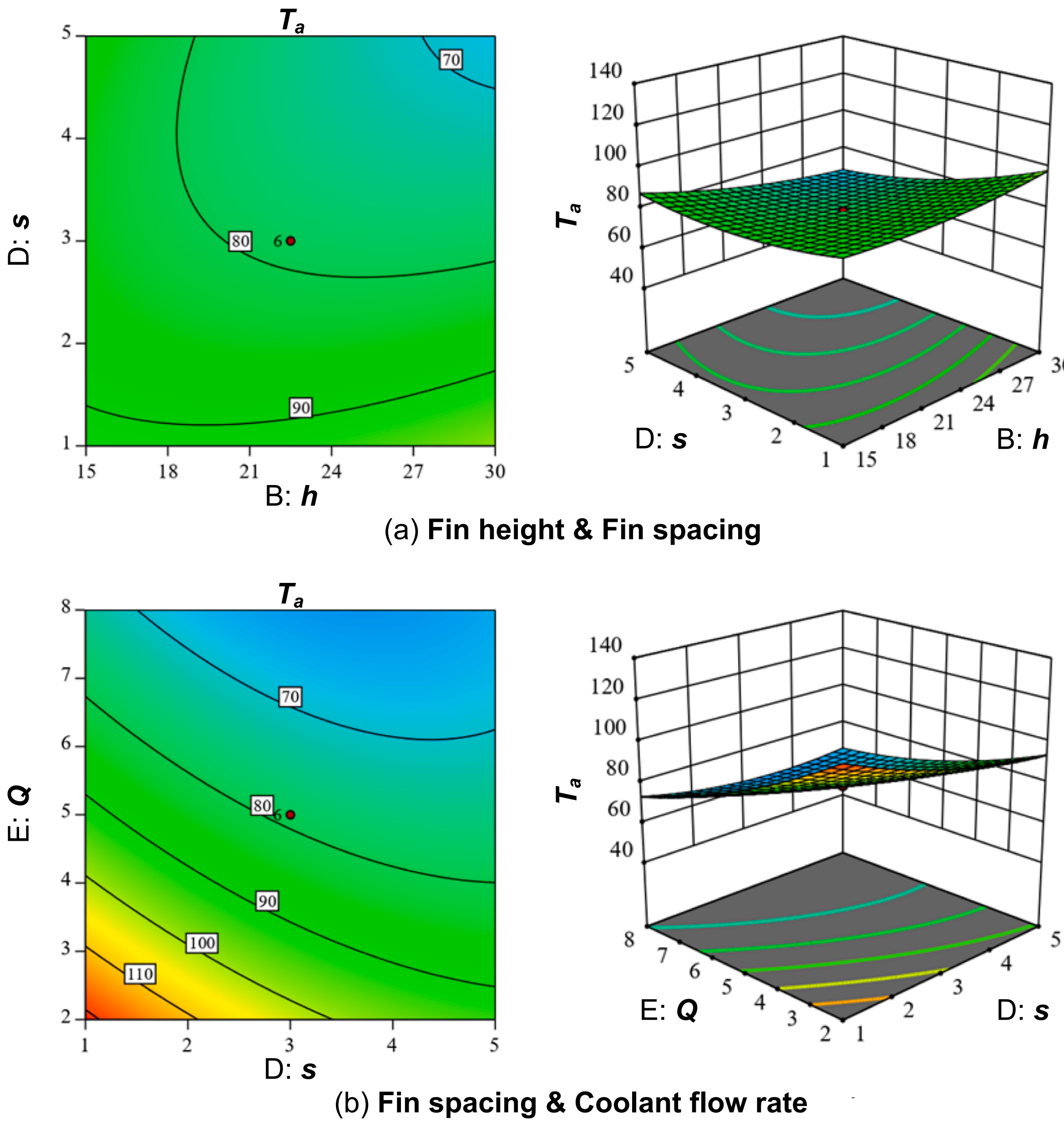

Fig. 7. Interaction effects on average CPUs temperature: (a) fin height and fin spacing. (b) Fin spacing and coolant flow rate.

significance of the model and the strength of interaction for each independent variable using key statistical indicators, including the p-value and F-value. The p-value serves as the primary criterion for assessing the significance of individual parameters, their interactions, and the model's overall validity. When the p-value is below the 0.05 significance level, it indicates that the associated model term has a statistically significant impact on the response variable. Conversely, p-values greater than 0.1 suggest that the term is not statistically significant and does not meaningfully contribute to the model. Additionally, a higher F-value indicates a greater degree of significance for the associated model term.

Among the independent parameters evaluated, Q exhibits the most significant influence on both the thermal and hydraulic performance of the immersion cooling system. The sensitivity analysis reveals the following order of influence on T_a: $Q > T_{in} > s > t > h > l$, indicating that the operating parameters of FC-40 as the coolant, particularly Q, play a more critical role than geometric parameters of HSs in governing overall cooling efficiency. For ΔT_m, the order of influence is: $Q > h > l > s > t > T_{in}$. This emphasizes the key role of Q in achieving thermal uniformity across both CPUs. Enhancing Q in conjunction with optimized HS geometries contributes to a more uniform temperature distribution and improved system reliability. In terms of P_P, the ranking is: $Q > h > s > l > t > T_{in}$, again highlighting Q as the primary contributor to hydraulic performance and energy consumption. Finally, for ΔT_c, the order of influence is: $Q > t > h > l > s > T_{in}$, further confirming the dominant role of Q in both primary and secondary energy metrics. The final quadratic regression equations, which describe the influence of the independent variables on T_a, ΔT_m, P_P, and ΔT_c in relation to the design parameters, are summarized in Appendix B as Eqs. B-1 through B-4. These equations effectively capture the system's behavior across the actual levels of the evaluated factors, demonstrating a strong fit to the simulations results.

As illustrated in Fig. 7a, increasing both h and s results in a reduction in T_a during SPIC. The increase in h enhances the total heat transfer surface area, thereby improving convective heat transfer. This facilitates greater heat absorption from the CPUs into the surrounding fluid, leading to a noticeable decrease in their operating temperature. For instance, when Q is maintained at 2 lpm, doubling h from 15 mm to 30 mm reduces the temperature of CPU 1 from 92.8 °C to 74.6 °C, representing a 19.6 % decrease. Similarly, the temperature of CPU 2 decreases

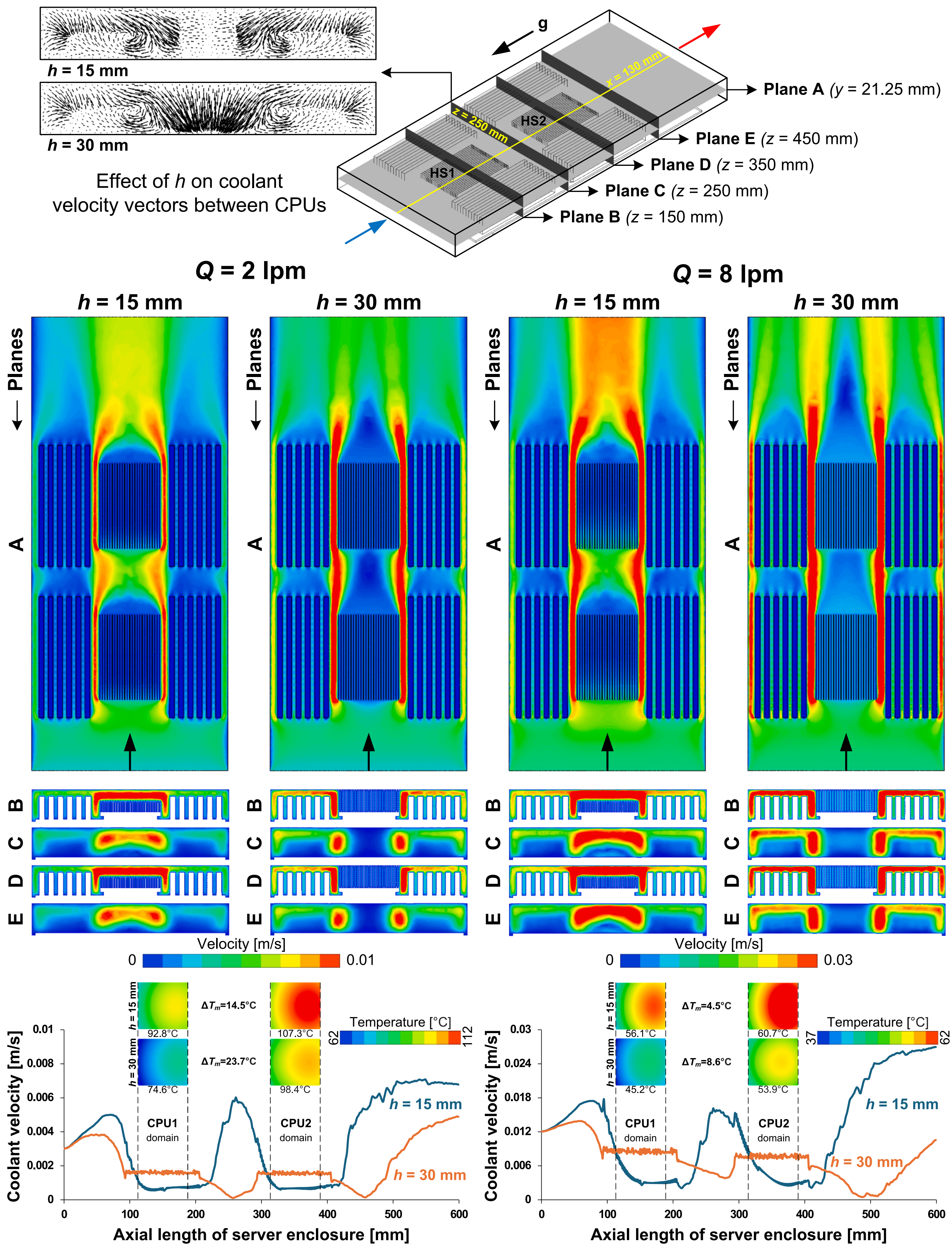

Fig. 8. Effects of fin height ($h = 15$ and 30 mm) and coolant flow rate ($Q = 2$ and 8 lpm) on the flow features of FC-40 within the data center server enclosure (with constant parameters: $l = 112.5$ mm, $t = 0.3$ mm, $s = 3$ mm, and $T_{in} = 20$ °C).

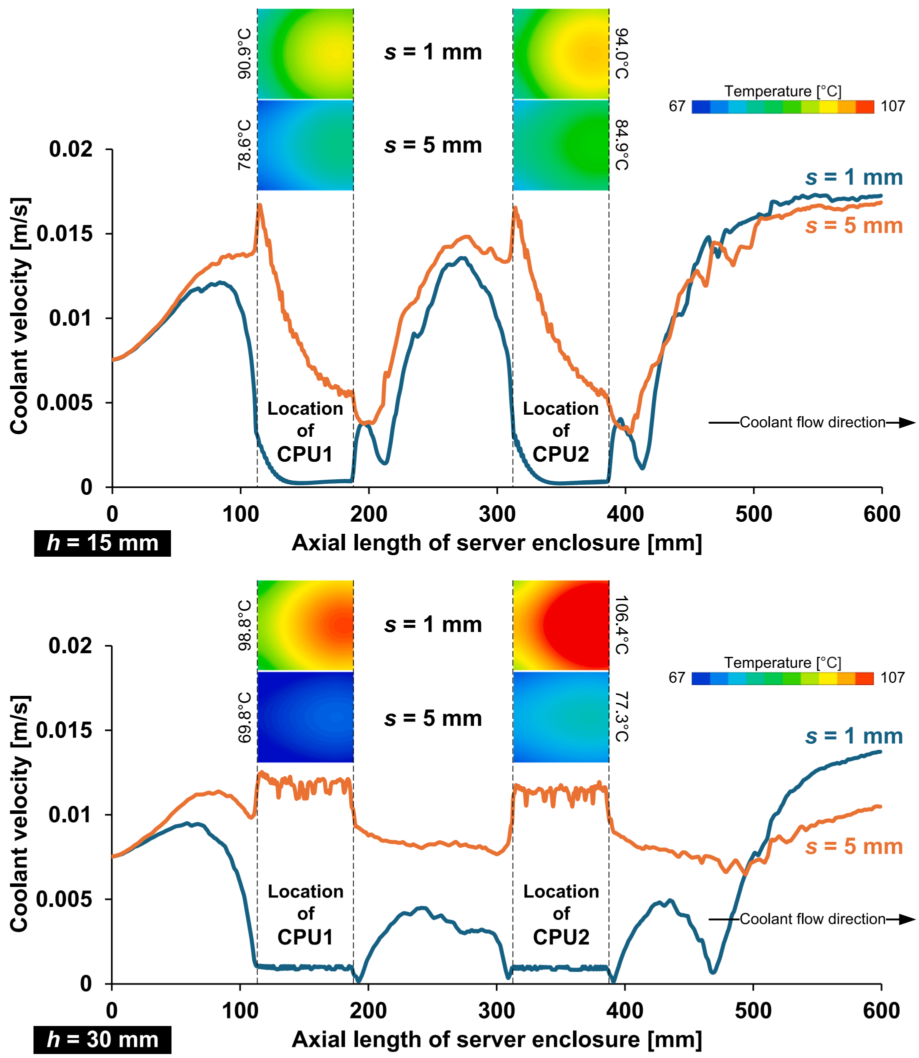

Fig. 9. Combined effects of fin height and fin spacing on the velocity distribution of FC-40 and the temperature profile of heat sinks within the data center server enclosure (with constant parameters: l = 75 mm, t = 0.3 mm, Q = 5 lpm, and T_{in} = 30 °C).

from 107.3 °C to 98.4 °C, corresponding to an 8.3 % reduction. However, excessively tall fins may introduce drawbacks, particularly in systems using viscous coolants like FC-40. As shown in Fig. 8, a greater h can increase flow resistance and lead to the formation of low-velocity or even stagnant regions between CUPs, especially at a low Q. As indicated by the velocity vectors, downstream of CPU 1, the elevated resistance causes a loss of coolant momentum, promoting secondary flows and formation of recirculation zones. These flow disruptions impede effective heat removal from CPU 2 and may lead to thermal stratification or the formation of localized hot spots in the second half of the server. This phenomenon significantly impacts thermal performance, as evidenced by a 63.4 % increase in ΔT_m at Q = 2 lpm when h is increased from 15 mm to 30 mm. Notably, this discrepancy further escalates to 91.1 % at Q = 8 lpm, highlighting the compounded influence of h and flow dynamics on overall cooling efficiency. However, the increased obstruction of the flow path caused by taller fins enhances coolant velocity around the DIMMs, thereby improving their cooling performance. For instance, at Q = 2 lpm, implementing 30 mm fins on the HSs results in a 12.8 % reduction in the average DIMMs temperature (T_d). Nevertheless, in the context of server thermal management, maintaining effective CPU cooling and ensuring temperature uniformity across CPUs remains the primary concern. Moreover, taller fins lead to heavier heat sinks, which may raise manufacturing complexity and cost.

In contrast, increasing s, which simultaneously reduces the number of fins, can partially offset the additional material usage and weight associated with taller fins. A larger s also widens the flow channels for the FC-40 coolant through the HSs, thereby reducing the risk of flow blockage and enhancing overall heat dissipation. This is achieved by facilitating better coolant access to the heated surfaces of the CPUs, thereby improving convective heat transfer. Although a smaller s can be employed at high Q due to enhanced forced convection, improper optimization of s may lead to significant drawbacks. Specifically, overly narrow spacing substantially increases pressure drop, which not only elevates P_P but also deteriorates coolant distribution. For example, at s = 1 mm (with constant parameters: l = 75 mm, h = 22.5 mm, t = 0.3 mm, and T_{in} = 30 °C), increasing Q from 2 to 8 lpm can reduce T_a by 41.7 %, but it also results in a P_P rise of up to 24-fold. Therefore, an optimal configuration typically involves a medium to wide s combined with a sufficiently high Q, ensuring a balanced trade-off between heat transfer surface area and fluid mobility.

As illustrated in Fig. 9, the elevated fluid velocity in channels with a larger s improves coolant circulation, helping to suppress the development of localized hotspots on CPU surfaces. This leads to more effective and uniform thermal management across the system. Furthermore, a larger s simplifies the manufacturing process and facilitates easier maintenance operations, such as cleaning and servicing of the HSs. Fig. 7b further confirms that simultaneously increasing both s and Q results in a lower T_a. Kim et al. [52] also recommended increasing s to

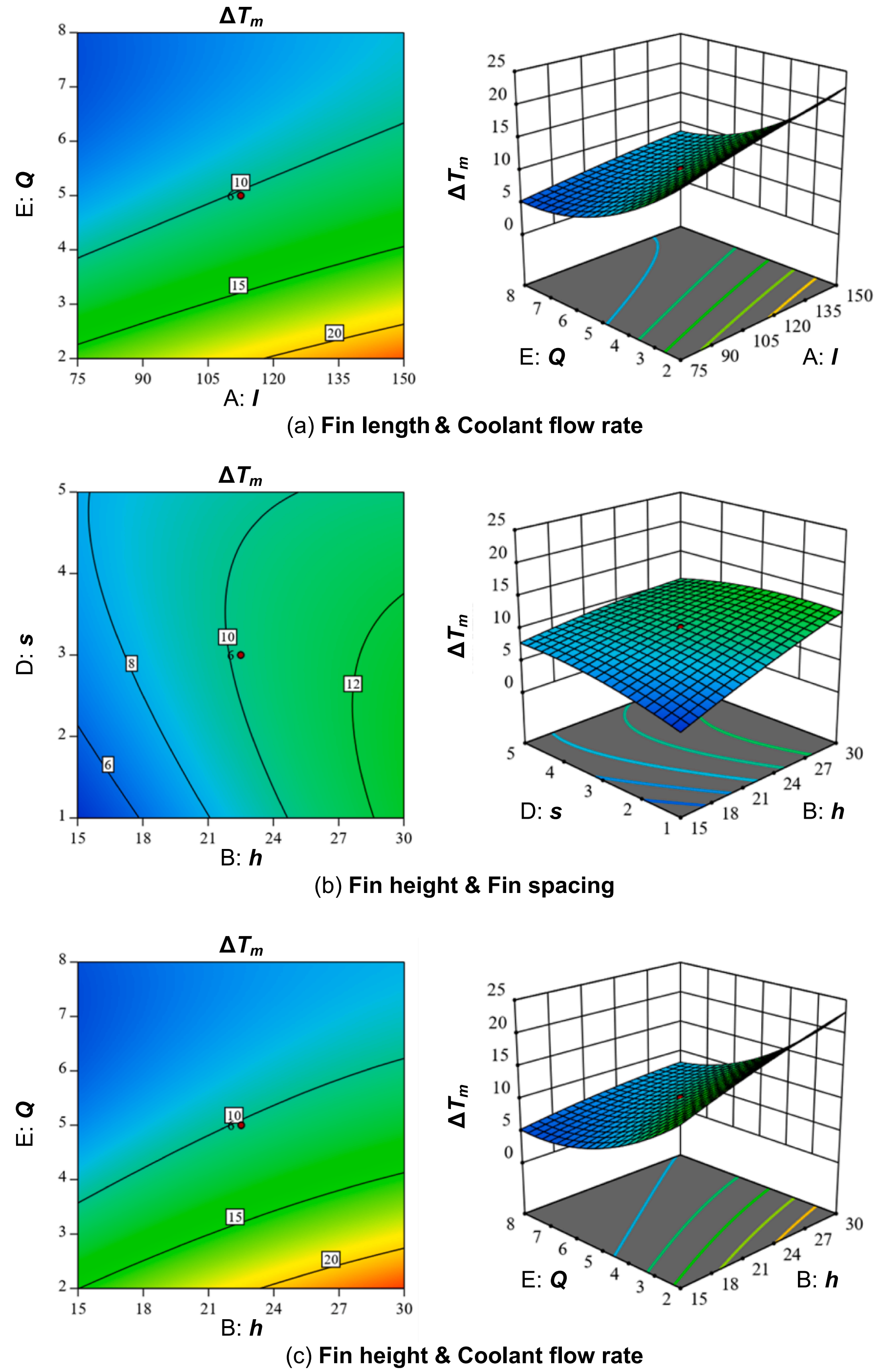

Fig. 10. Interaction effects on maximum CPUs temperature difference: (a) fin length and coolant flow rate. (b) Fin height and fin spacing. (c) Fin height and coolant flow rate.

facilitate the flow of high-viscosity dielectric fluids, thereby enhancing the convective heat transfer coefficient. Similarly, Wu et al. [45] found that increasing t, which directly reduces s for a fixed number of fins, does not provide significant thermal benefits beyond a thickness of 0.5 mm.

Although l has the least influence on T_a among the independent parameters, it can significantly affect ΔT_m and P_P, and ΔT_c. This influence is evident in the interaction between l and Q, as illustrated in Fig. 10a. The results indicate that reducing l in combination with increasing Q leads to a substantial decrease in ΔT_m. To evaluate the effect of l on the cooling performance of the system, the variations of key parameters at the minimum and maximum levels of l and Q across the transverse plane ($x = 130$ mm) are presented in Fig. 11. The influence of l on coolant flow characteristics within the server enclosure is clearly observable through the velocity contours. Although employing longer fins increases the heat transfer area and extends the contact time between the coolant and CPU surfaces, thereby reducing CPU temperatures, it also amplifies the temperature difference between CPU 1 and CPU 2, resulting in higher ΔT_m values. This occurs because the first HS with a greater l absorbs more heat from CPU 1, thereby increasing the temperature of the coolant as it exits the HS. Consequently, the cooling effectiveness for CPU 2 diminishes, particularly under low Q conditions. Moreover, the effect of l on the flow pattern downstream of CPU 1 plays a crucial role in the thermal performance of CPU 2. The velocity vectors indicate that reducing l causes the streamlines to deflect more effectively

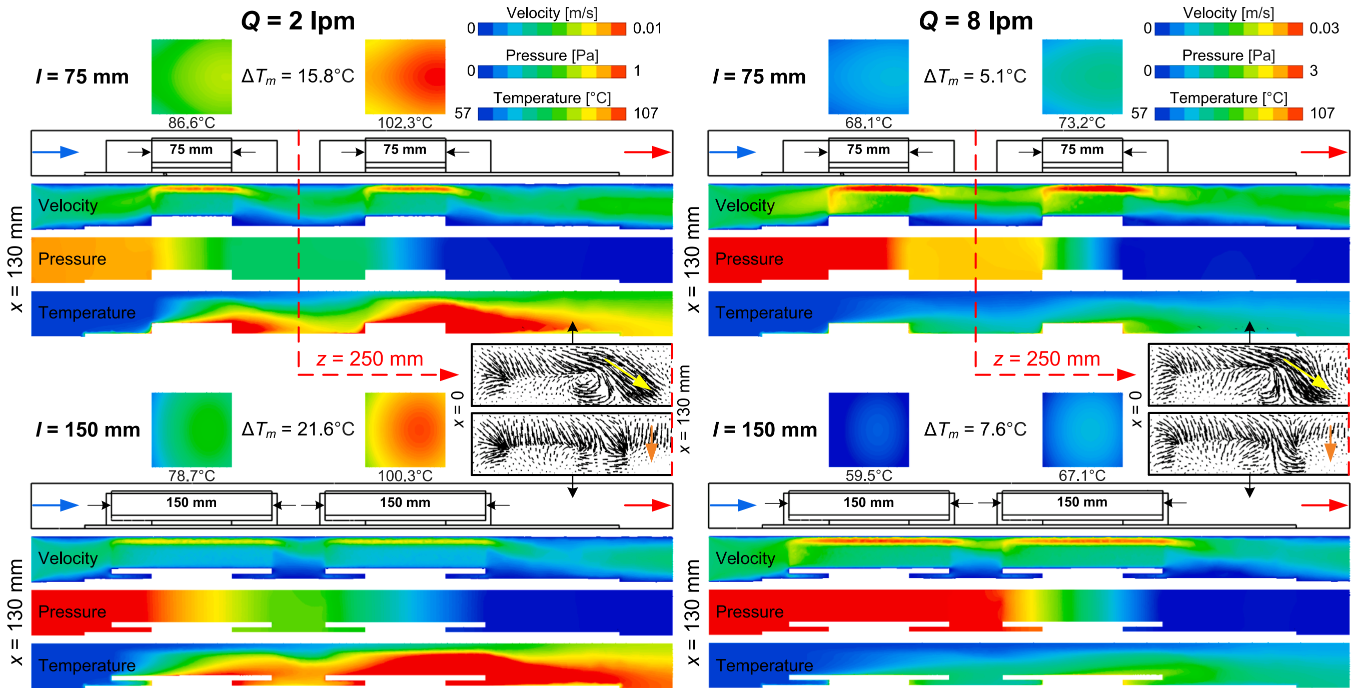

Fig. 11. Effects of fin length (l = 75 and 150 mm) and coolant flow rate (Q = 2 and 8 lpm) on the velocity, pressure, and temperature distributions of FC-40 within the data center server enclosure (with constant parameters: h = 22.5 mm, t = 0.3 mm, s = 5 mm, and T_{in} = 30 °C).

toward CPU 2. This behavior resembles jet flow, which enhances the cooling performance of CPU 2 by directing more coolant toward it.

The pressure contours confirm that longer channels increase flow resistance, resulting in a higher pressure drop and greater P_P. For instance, under the conditions examined in Fig. 11, doubling l increases P_P by 29.9 % at $Q = 2$ lpm and by 23.9 % at $Q = 8$ lpm. Additionally, the temperature contours indicate that HSs with longer l result in higher outlet coolant temperatures. This, in turn, elevates secondary power consumption required to restore the coolant temperature and return it to the cooling cycle. From a thermal and mechanical design perspective, increasing l also entails added weight, material usage, and overall system size, without providing a proportionate improvement in cooling performance. Therefore, considering all key performance parameters, it can be predicted that designing the HSs length to match the CPUs length may enhance cooling performance in SPIC systems.

In contrast to the interaction observed between h and s in Fig. 7a, where increasing both parameters results in lower T_a values, the findings in Fig. 10b indicate that decreasing these parameters improves temperature uniformity between CPUs and leads to a reduction in ΔT_m values. Further evidence of enhanced uniformity with lower h values is shown in the interaction between h and Q, as depicted in Fig. 10c. This figure demonstrates that selecting the minimum h in combination with a sufficiently high Q enables the attainment of the target temperature uniformity between CPUs (i.e., $\Delta T_m < 5$ °C). Moreover, this configuration significantly contributes to a reduction in P_P, primarily by minimizing flow resistance due to lower obstruction and decreasing pressure drop at reduced Q. This trend is further confirmed in Fig. 12a. A similar analysis, based on Fig. 12b, can be carried out to examine the interaction between l and Q. As previously discussed, l exhibits minimal influence on T_a and has a slightly adverse effect on ΔT_m. Consequently, selecting the minimum value of l reduces the pressure drop typically associated with longer flow paths through narrow channels, thereby enabling the application of higher Q values under a constant P_P condition. Regarding Fig. 12c, the interaction between s and Q on P_P is of a similar order of magnitude to that of the interaction between l and Q, yet it is substantially weaker than the interaction between h and Q. This observation is further supported by the F-values reported in Tables A-3, where the interaction terms AE and DE have F-values of 25.47 and 37.47, respectively, compared to a significantly higher value of 311.37 for the interaction term BE.

The results presented in Fig. 13a indicate that a reduction in h positively impacts decreasing ΔT_c, thereby lowering the secondary energy required to restore the coolant temperature. Furthermore, when this reduction in h is combined with an increase in t, the minimum ΔT_c values are achieved. However, as shown in Fig. 13b, the influence of t is more pronounced at lower values of Q. As Q increases beyond 6 lpm, the effect of t on ΔT_c diminishes, leading to a convergence in ΔT_c values, with the lowest values being attained regardless of further increases in t.

The optimization criteria aim to minimize all response parameters listed in Table 5, namely T_a, ΔT_m, P_P, and ΔT_c, while ensuring that all independent variables remain within their specified bounds. Among the items evaluated in Table 6, Case No. 1, characterized by dimensions of l = 75 mm, h = 15.01 mm, t = 0.5 mm, s = 3.06 mm, Q = 6.62 lpm, and T_{in} = 20.06 °C, is identified as the optimal case. To develop a specific and practically manufacturable geometric model for the HSs, and to establish controllable operating conditions, all independent variables are rounded to appropriate levels of precision. It should be noted that due to the substantial impact of server layout on SPIC, the developed model is specifically applicable to the server configuration considered in this study. Its accuracy may be affected by changes in the arrangement of CPUs and DIMMs or by the inclusion of additional components within the server chassis. Nevertheless, the final optimized model is utilized in the following sections to investigate the effects of TDP and coolant type.

4.2. Analysis under different TDP scenarios

In recent years, the rapid advancement of AI technologies along with the increasing demand for high-speed processing has significantly raised the TDP requirements of CPUs used in DC servers. Despite this upward trend, most existing research on SPIC have primarily focused on CPUs with low to medium TDP levels [27]. This section investigates the performance of the optimized heat sink model (l = 75 mm, h = 15 mm, t = 0.5 mm, and s = 3 mm) and its corresponding operating conditions (Q = 7 lpm and T_{in} = 20 °C) for SPIC over a broad range of TDP values, from

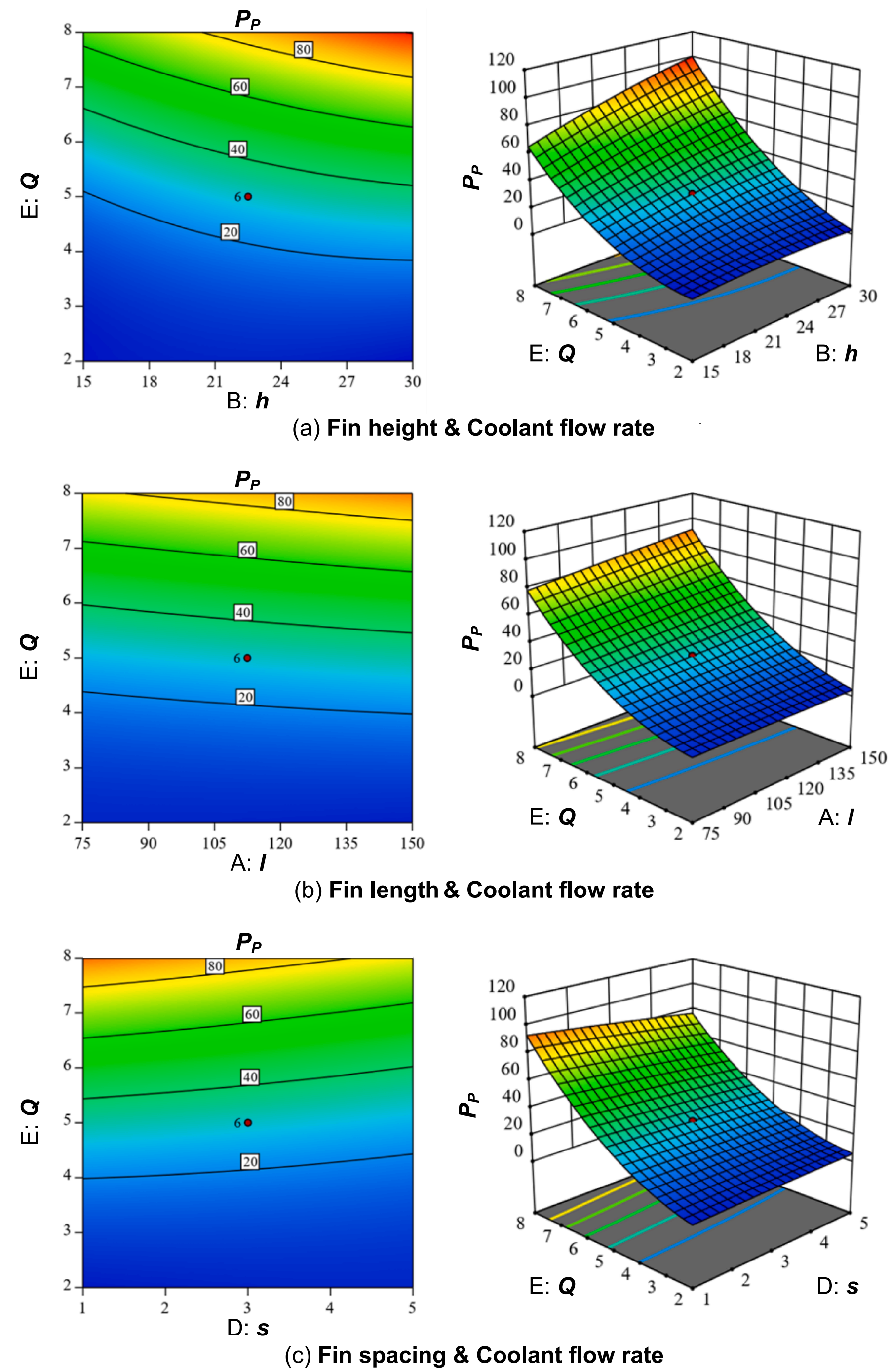

Fig. 12. Interaction effects on coolant pumping power ($\times 10^{-5}$): (a) fin height and coolant flow rate. (b) Fin length and coolant flow rate, (c) fin spacing and coolant flow rate.

100 W to 800 W. These values are representative of typical loads in DC servers and extend to the upper limits reported in previous studies [27]. The objective is to evaluate whether SPIC, in conjunction with the proposed heat sink design and optimized operational settings, can effectively meet the thermal management needs of next-generation DC CPUs operating under higher TDP conditions.

Fig. 14 illustrates the variations in thermal resistance (R), T_a, ΔT_m, and ΔT_c with increasing TDP. The results demonstrate a consistent decrease in R as TDP increases for both CPUs. This trend is primarily attributed to changes in the physical properties of the coolant, most notably, its μ, which decreases by approximately 16.7 % as TDP increases from 100 to 800 W. This finding contrasts with the typical behavior observed in air-cooled systems, where R generally remains relatively constant despite changes in TDP [43]. Notably, the reduction in R with increasing TDP is more pronounced for CPU 2 compared to CPU 1. Within the examined TDP range, CPU 1 exhibits a 7.7 % decrease in R, whereas CPU 2 shows a more substantial reduction of 9.8 %. This difference is attributed to the enhanced convective performance of the coolant flowing through CPU 2. The reduction in μ the coolant facilitates more effective convective heat transfer, which in turn leads to improved cooling performance and a more pronounced decrease in R.

The findings indicate that thermal management of DC servers using the proposed cooling system can be effectively maintained for silicon-based CPUs with TDP values up to 600 W, ensuring their temperatures

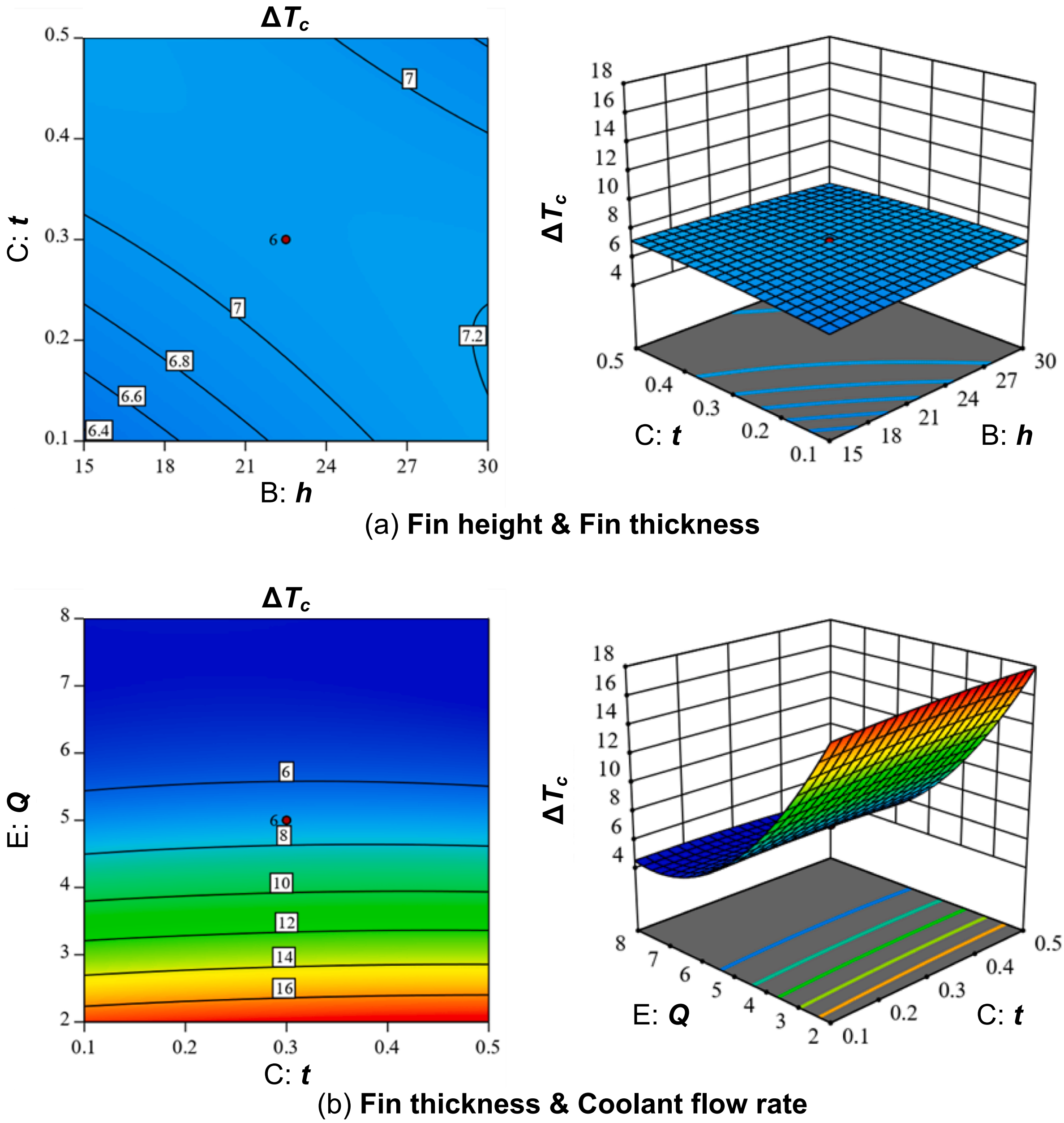

Fig. 13. Interaction effects on coolant temperature difference: (a) fin height and fin thickness. (b) Fin thickness and coolant flow rate.

Table 6
Optimized cases identified through RSM-BBD analysis.

Number	A: l [mm]	B: h [mm]	C: t [mm]	D: s [mm]	E: Q [lpm]	F: T_{in} [°C]	Ta [°C]	ΔT_m [°C]	$P_P \times 10^{-5}$ [W]	ΔT_c [°C]	Desirability
1	75.00	15.01	0.50	3.06	6.62	20.06	57.82	3.12	33.57	4.99	0.91
2	75.00	15.37	0.50	2.89	6.49	20.00	56.61	3.10	36.69	4.87	0.90
3	75.00	15.00	0.50	3.09	6.57	20.13	57.46	3.12	34.70	4.92	0.90
4	75.00	15.00	0.50	3.15	6.68	20.08	56.86	3.15	36.02	4.83	0.90
5	75.00	15.08	0.50	2.86	6.76	20.00	56.23	2.93	38.18	4.77	0.90
6	75.72	15.44	0.50	2.82	6.56	20.12	57.12	3.12	36.23	4.92	0.90
7	75.00	15.63	0.50	2.76	6.48	20.10	57.39	3.12	35.54	4.99	0.90
8	75.15	15.02	0.50	2.89	6.88	20.02	55.69	2.94	39.84	4.69	0.89
9	75.00	15.02	0.49	3.09	6.64	20.05	57.45	3.12	35.60	4.87	0.89
10	75.70	15.00	0.50	3.15	6.44	20.13	58.20	3.24	32.76	5.04	0.89

remain within the permissible operating range (i.e., below 85 °C) [64]. Moreover, for servers utilizing chips with higher thermal stability, such as gallium nitride (GaN)-based devices commonly employed in high-power and specialized high-frequency applications, the proposed cooling system remains effective due to these chips' elevated permissible operating temperatures, which can exceed 100 °C. The ΔT_m values obtained for TDP levels below 600 W remain within the acceptable range, consistently staying below 5 °C. This observation highlights both the

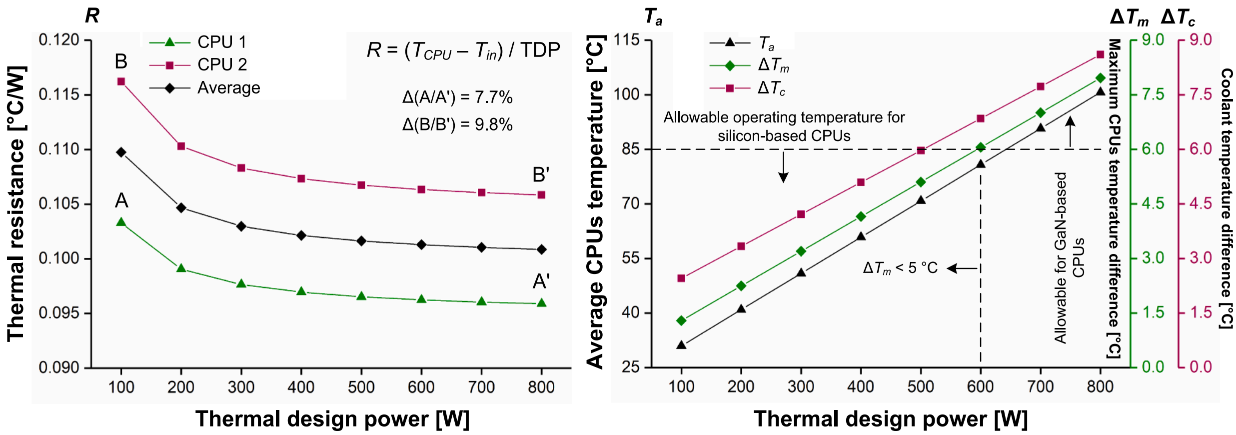

Fig. 14. Variations in thermal resistance, average CPU temperature, maximum CPUs temperature difference, and coolant temperature difference as a function of thermal design power.

thermal stability of the system and the notable uniformity in temperature distribution among the CPUs within the server. Notably, for every 100 W increase in TDP, the ΔT_m increases by approximately 1 °C. Specifically, at a TDP of 100 W, ΔT_m is measured at 1.29 °C, rising to 7.96 °C at 800 W. As expected, the coolant temperature increases with higher TDP values due to greater heat absorption. Nevertheless, across the entire studied TDP range, the ΔT_c remains below 10 °C, indicating a substantial reduction in the energy required during the coolant cooling cycle to restore its temperature. For example, at a TDP of 500 W, ΔT_c is 5.97 °C, which closely corresponds with experimental results reported by Shrigondekar et al. [43] for FC-40 fluid.

4.3. Performance comparison of various coolants

Research indicates that the type of coolant plays a critical role in the thermal management of DC servers. For instance, studies have demonstrated that in immersion cooling systems, replacing air with liquid coolants can significantly improve heat dissipation, reducing chip temperatures by more than 20 % [65]. This section presents a performance evaluation of various dielectric liquids by comparing their thermal and hydraulic characteristics. The objective is to identify the most suitable coolant for SPIC, based on critical performance metrics. The fluids assessed in this study fall into four main categories: fluorocarbons (FC-40, FC-74, FC-75, FC-77, and FC-3283), hydrofluoroethers (HFE-7200, HFE-7300, HFE-7500, and HFE-7600), engineered coolants (Novec^{TM} 7200) [64], and oils (Mineral, Synthetic, and Silicone) []. To ensure a fair comparison under identical conditions, the TDP of the CPUs is assumed to be 400 W, while that of the DIMMs is set to 15 W. Furthermore, the geometric parameters of the heat sink and the operating conditions are taken at their optimized values.

Fig. 15 illustrates the impact of coolant type on key performance parameters. The results clearly indicate that the selection of coolant significantly influences on the thermal and hydraulic behavior of the SPIC process. Among the tested coolants, synthetic oil yields the lowest T_a of 57.8 °C, followed closely by Novec^{TM} 7200, which achieved a T_a of 58.3 °C. In addition, Novec^{TM} 7200 exhibits the lowest P_P, indicating its superior hydraulic performance. The enhanced performance of Novec^{TM} 7200 can be attributed to the unique thermophysical properties of Novec^{TM} 7200, which include the lowest μ and the highest λ, excluding coolants within the oil category, among all fluids evaluated. The findings suggest that coolants with lower μ and higher λ generally lead to improved thermal and hydraulic performance across nearly all coolant categories. This conclusion aligns with prior research [66], which has shown that μ and λ have a more pronounced impact on the cooling efficiency of SPIC systems than ρ and c_p.

Within the fluorocarbon category, FC-74 exhibits the lowest μ and a moderate λ, which accounts for its relatively low T_a and P_P. Notably, FC-74 also results in higher values of ΔT_m and ΔT_c compared to the other fluorocarbons in the same group. In the hydrofluoroether category, HFE-7200 demonstrates superior performance, primarily due to its low μ and high λ. Although it yields a higher ΔT_m compared to the other coolants in this group, its ΔT_c remains nearly equivalent to that of the others. This suggests that the secondary energy required for thermal recovery is comparable across the fluids. Additionally, HFE-7000′s low ρ further contributes to its favorable hydraulic characteristics, making it the most efficient coolant among the hydrofluoroethers evaluated. The results also indicate that, for the engineered coolant of Novec^{TM} 7200, the influence of μ on cooling performance is more pronounced than that of λ. Choi et al. [67] previously underscored the critical role of μ in the performance of engineered coolants used in SPIC systems. In the oil category, the influence of μ is significantly more pronounced than that of other thermophysical properties. Among the evaluated oils, synthetic oil exhibits the best overall performance, primarily due to its lowest μ. Specifically, the use of synthetic oil results in a reduction of T_a by 3.9 °C and 2.7 °C compared to mineral and silicone oils, respectively. Additionally, it improves ΔT_m and ΔT_c by approximately 1 °C, while lowering P_P by a factor of about 1.58 compared to mineral oil and more than 7.18 times relative to silicone oil. The sequential trends demonstrating the influence of thermophysical properties on key performance parameters for each category of coolant are identified through sensitivity analysis and are summarized in Fig. 15.

However, the effectiveness of different coolants in DIMMs cooling varies depending on their thermophysical properties and flow conditions. The results indicate that oil coolants are generally more effective in achieving lower T_d. For instance, when compared to FC-40, silicone oil reduces the average T_d by up to 3.5 °C while also decreasing the P_P by approximately 20 %. This enhanced performance can be attributed to the less restricted flow paths between DIMMs, which are typically spaced further apart than the fins in the HS (s). This configuration allows for more efficient convective heat transfer, particularly when using oil coolants with relatively high λ and c_p. Moreover, despite the inherently higher μ of oil coolants, the design of the DIMMs region does not impose significant flow resistance, minimizing the negative impact of μ on hydraulic performance. Additionally, the increased cross-sectional area for coolant flow around the DIMMs reduces fluid velocity, increasing the relative contribution of natural convection compared to forced convection. Under such conditions, the benefits of higher λ become more prominent, further enhancing the cooling efficiency of oil coolants in this region. Consequently, in addition to providing more effective DIMM cooling compared to other coolants, oil coolants present several advantages that make them highly suitable for SPIC applications. These benefits include low toxicity, non-corrosiveness, environmental

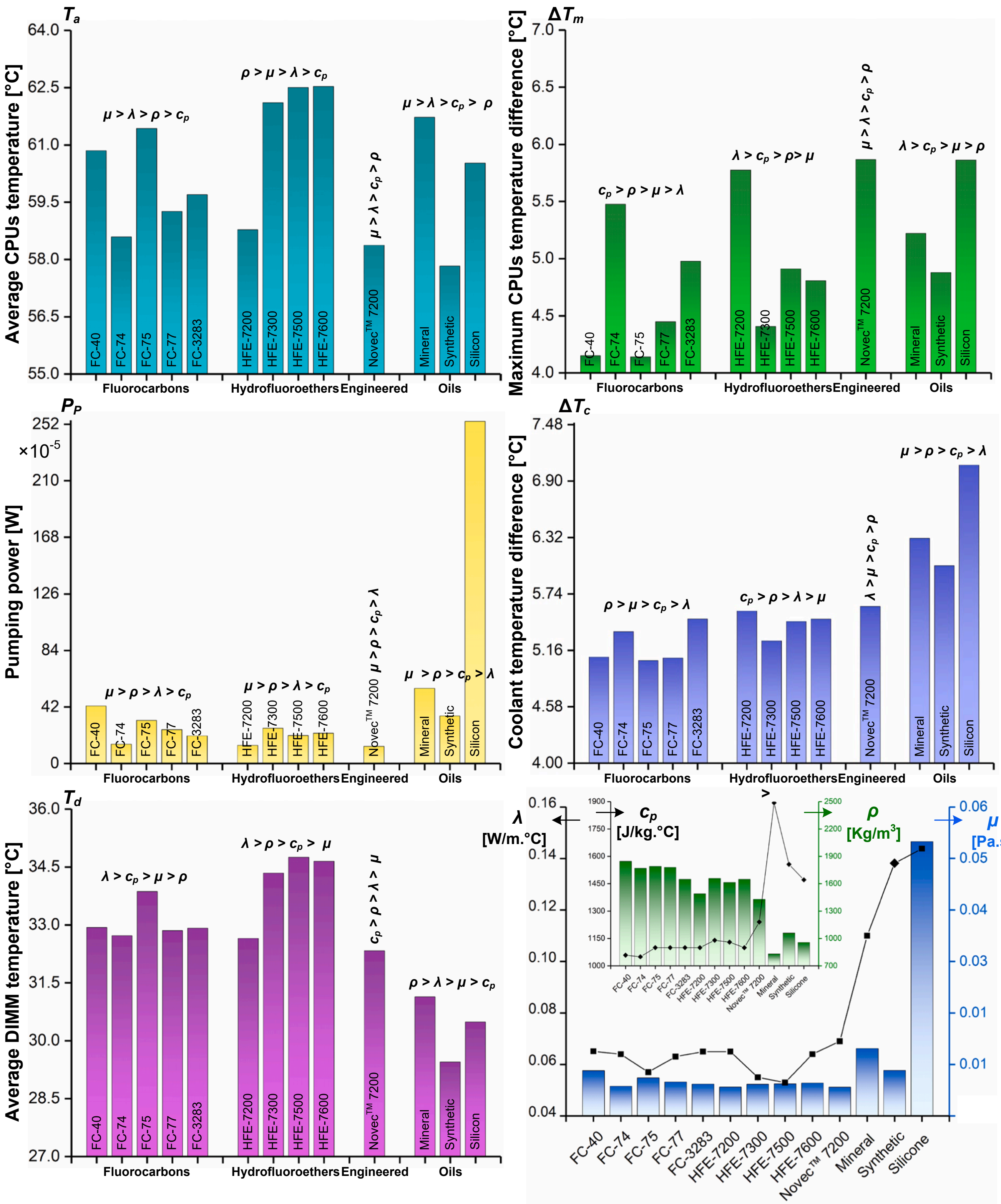

Fig. 15. Effects of coolant type on average CPUs temperature, maximum CPUs temperature difference, pumping power, coolant temperature difference, and average DIMMs temperature.

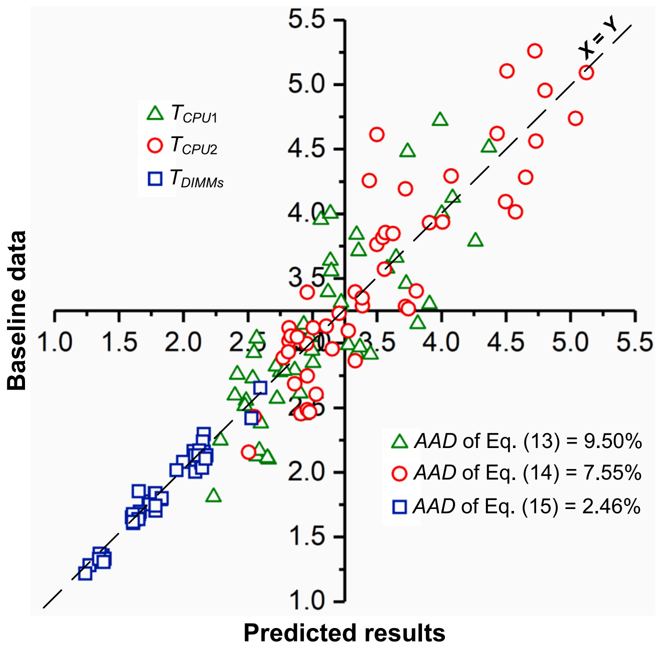

Fig. 16. Comparison of baseline data with predicted results from the developed correlations.

compatibility, cost-effectiveness, and relatively high boiling points. Nevertheless, to fully realize their potential and optimize performance, further dedicated investigations are required.

4.4. Proposed correlations

To enhance the practical applicability of this research, dimensionless correlations are developed to predict the temperatures of key components within DC servers cooled using FC-40. These correlations are formulated based on independent variables, allowing for the direct estimation of the temperatures of CPU 1, CPU 2, and DIMMs. The significance of these correlations lies in their ability to estimate component temperatures as a function of HS geometric parameters, coolant operating conditions and thermophysical properties. The derived correlations are presented as follows:

$$\frac{T_{CPU1}}{T_{amb}} = 13.7109Re^{-0.2769}\left(\frac{T_{in}}{T_{amb}}\right)^{0.3795}\left(\frac{h}{l}\right)^{-0.0347}\left(\frac{t}{s}\right)^{0.0424} \quad (9)$$

$$\frac{T_{CPU2}}{T_{amb}} = 21.8058Re^{-0.3384}\left(\frac{T_{in}}{T_{amb}}\right)^{0.3304}\left(\frac{h}{l}\right)^{-0.0241}\left(\frac{t}{s}\right)^{0.0303} \quad (10)$$

$$\frac{T_{DIMMs}}{T_{amb}} = 4.2459Re^{-0.1886}\left(\frac{T_{in}}{T_{amb}}\right)^{0.6513}\left(\frac{h}{l}\right)^{-0.0388}\left(\frac{t}{s}\right)^{-0.0023} \quad (11)$$

where Re denotes the Reynolds number, which captures the combined effects of the coolant's properties and flow rate. T_{amb} represents the ambient temperature, set at 25 °C, and is used as the reference value to nondimensionalize the temperature variables. To validate the accuracy of the developed correlations, the baseline data are compared with the values predicted by Eqs. (13), (14), and (15) in Fig. 16. The results indicate that the majority of the baseline data points fall within the 90 % confidence interval of the predicted values. Additionally, the Average Absolute Deviation (AAD) for each parameter is calculated using the following equation, with the results also illustrated in the figure. The calculated AAD values are 9.50 % for T_{CPU1}, 7.55 % for T_{CPU2}, and 2.46 % for T_{DIMMs}.

$$AAD(\%) = \frac{|T_{correlation} - T_{baseline}|}{T_{baseline}} \times 100 \quad (12)$$

5. Conclusion and future research directions

This study focuses on the optimization of heat sink geometries and coolant operating conditions in a dual-CPU server equipped with 24 DIMMs immersed in FC-40 dielectric fluid. By employing response surface methodology based on a Box-Behnken design (RSM-BBD), the influence of critical geometric and operational parameters is systematically evaluated and optimized. The main findings of this research are summarized as follows:

- New correlations are developed to predict the thermophysical properties of FC-40 (thermal conductivity, specific heat capacity, density, and viscosity) with less than 1 % deviation across the temperature range of 0 to 100 °C, supporting more precise thermal modeling.
- The sensitivity analysis reveals that the coolant flow rate (Q) is the most influential factor for all performance metrics, particularly in reducing average CPUs temperature (T_a). Geometric parameters (l, h, t, and s) show stronger effects than inlet temperature (T_{in}) on
- maximum CPUs temperature difference (ΔT_m), coolant pumping power (P_P), and coolant temperature difference (ΔT_c).
- The optimal case ($l = 75$ mm, $h = 15$ mm, $t = 0.5$ mm, $s = 3$ mm, $Q = 7$ lpm, and $T_{in} = 20$ °C) maintains CPU temperatures below 60 °C at a thermal design power of 400 W, while ensuring excellent temperature uniformity ($\Delta T_m < 5$ °C) and low P_P and ΔT_c values. The optimized design maintains safe operating temperatures up to 600 W (silicon-based CPUs) and remains effective even at 800 W for advanced GaN-based processors.
- Among the tested coolants, Novec™ 7200 offers the best overall performance, achieving the lowest T_a of 56.3 °C and the lowest P_P of 12.8×10^{-5} W. Within the fluorocarbon category, FC-74 resulted in the lowest T_a of 58.5 °C, while FC-40 and FC-75 exhibited the most uniform temperature distributions, with a minimum ΔT_m of approximately 4.1 °C.
- The proposed correlations accurately predict CPU and DIMM temperatures, with average absolute deviations of 9.50 % for T_{CPU1}, 7.55 % for T_{CPU2}, and 2.46 % for T_{DIMMs}.

The results confirm that the optimized thermal management model provides effective cooling performance for data center servers, even under high CPU thermal loads. To facilitate practical implementation, the proposed heat sinks are currently being fabricated and evaluated at the laboratory scale. Future research should focus on exploring advanced heat sink configurations tailored for both single-phase and two-phase immersion cooling, optimized based on server architecture and coolant properties. Additionally, integrating the optimized designs with passive (e.g., internal baffles, heat pipes) or active (e.g., liquid jets, micropumps) enhancement techniques may further improve thermal performance and address the growing demands of next-generation high-performance computing systems.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to modify the English of paper. After using this service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

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CRediT authorship contribution statement

M. Khoshvaght-Aliabadi: Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **P. Ghodrati:** Investigation, Data curation, Writing – original draft. **A. Nasrolahzadeh:** Validation, Investigation. **Y.T. Kang:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Analysis of variance (ANOVA) Tables

This appendix presents the ANOVA results for the key metrics analyzed in this study:
Table A-1: ANOVA for average CPUs temperature (T_a).
Table A-2: ANOVA for maximum CPU temperature difference (ΔT_m).
Table A-3: ANOVA for coolant pumping power (P_P).
Table A-4: ANOVA for coolant temperature difference (ΔT_c).

Table A-1
Analysis of variance for average CPUs temperature (T_a).

Source	Sum of squares	df	Mean square	F-value	p-value	
Model	15641.20	27	579.30	30.70	< 0.0001	Significant
A-l	68.87	1	68.87	3.65	0.0671	
B-h	245.93	1	245.93	13.03	0.0013	
C-t	708.44	1	708.44	37.55	< 0.0001	
D-s	1833.78	1	1833.78	97.19	< 0.0001	
E-Q	8700.52	1	8700.52	461.13	< 0.0001	
F-T_{in}	2400.26	1	2400.26	127.21	< 0.0001	
AB	11.68	1	11.68	0.6193	0.4384	
AC	8.63	1	8.63	0.4575	0.5048	
AD	10.20	1	10.20	0.5404	0.4688	
AE	12.20	1	12.20	0.6468	0.4285	
AF	1.037E-06	1	1.037E-06	5.495E-08	0.9998	
BC	0.5349	1	0.5349	0.0283	0.8676	
BD	320.66	1	320.66	17.00	0.0003	
BE	26.86	1	26.86	1.42	0.2436	
BF	1.815E-07	1	1.815E-07	9.620E-09	0.9999	
CD	18.01	1	18.01	0.9545	0.3376	
CE	0.1972	1	0.1972	0.0105	0.9194	
CF	6.528E-06	1	6.528E-06	3.460E-07	0.9995	
DE	218.58	1	218.58	11.58	0.0022	
DF	0.0000	1	0.0000	1.437E-06	0.9991	
EF	1.938E-07	1	1.938E-07	1.027E-08	0.9999	
A^2	6.14	1	6.14	0.3253	0.5733	
B^2	92.64	1	92.64	4.91	0.0357	
C^2	0.1880	1	0.1880	0.0100	0.9213	
D^2	253.84	1	253.84	13.45	0.0011	
E^2	255.77	1	255.77	13.56	0.0011	
F^2	92.12	1	92.12	4.88	0.0361	
Residual	490.57	26	18.87			
Lack of fit	490.57	21	23.36			
Pure error	0.0000	5	0.0000			
Cor total	16131.77	53				

Adj R-Squared = 0.9696, Pred R-Squared = 0.8411, Adequate Precision = 22.4666.

Table A-2
Analysis of variance for maximum CPUs temperature difference (ΔT_m).

Source	Sum of squares	df	Mean square	F-value	p-value	
Model	1471.18	27	54.49	66.15	< 0.0001	Significant
A-l	151.24	1	151.24	183.62	< 0.0001	
B-h	186.00	1	186.00	225.83	< 0.0001	
C-t	1.62	1	1.62	1.97	0.1727	
D-s	3.56	1	3.56	4.32	0.0476	
E-Q	916.84	1	916.84	1113.15	< 0.0001	
F-T_{in}	0.0001	1	0.0001	0.0001	0.9927	
AB	0.4163	1	0.4163	0.5054	0.4834	
AC	0.5149	1	0.5149	0.6252	0.4363	
AD	2.87	1	2.87	3.49	0.0732	
AE	5.54	1	5.54	6.72	0.0154	
AF	8.989E-06	1	8.989E-06	0.0000	0.9974	
BC	1.20	1	1.20	1.46	0.2380	
BD	17.25	1	17.25	20.94	0.0001	
BE	28.77	1	28.77	34.93	< 0.0001	
BF	9.730E-07	1	9.730E-07	1.181E-06	0.9991	
CD	1.08	1	1.08	1.31	0.2624	
CE	0.1768	1	0.1768	0.2146	0.6470	
CF	0.0001	1	0.0001	0.0001	0.9926	
DE	1.16	1	1.16	1.41	0.2456	
DF	0.0001	1	0.0001	0.0001	0.9907	
EF	9.180E-07	1	9.180E-07	1.115E-06	0.9992	
A^2	0.4570	1	0.4570	0.5549	0.4630	
B^2	1.92	1	1.92	2.34	0.1384	
C^2	0.2439	1	0.2439	0.2961	0.5910	
D^2	10.20	1	10.20	12.38	0.0016	
E^2	101.85	1	101.85	123.66	< 0.0001	
F^2	0.0236	1	0.0236	0.0286	0.8669	
Residual	21.41	26	0.8237			
Lack of fit	21.41	21	1.02			
Pure error	0.0000	5	0.0000			
Cor total	1492.60	53				

Adj R-Squared = 0.9856, Pred R-Squared = 0.925, Adequate Precision = 33.9352.

Table A-3
Analysis of variance for coolant pumping power (P_P).

Source	Sum of squares	df	Mean square	F-value	p-value	
Model	44930.52	27	1664.09	401.60	< 0.0001	Significant
A-l	239.58	1	239.58	57.82	< 0.0001	
B-h	1862.75	1	1862.75	449.54	< 0.0001	
C-t	0.1307	1	0.1307	0.0315	0.8604	
D-s	301.96	1	301.96	72.87	< 0.0001	
E-Q	38161.93	1	38161.93	9209.65	< 0.0001	
F-T_{in}	0.0000	1	0.0000	8.337E-06	0.9977	
AB	2.76	1	2.76	0.6671	0.4215	
AC	0.0129	1	0.0129	0.0031	0.9559	
AD	0.6534	1	0.6534	0.1577	0.6945	
AE	105.54	1	105.54	25.47	< 0.0001	
AF	1.494E-06	1	1.494E-06	3.606E-07	0.9995	
BC	1.64	1	1.64	0.3953	0.5350	
BD	16.57	1	16.57	4.00	0.0561	
BE	1290.24	1	1290.24	311.37	< 0.0001	
BF	1.842E-06	1	1.842E-06	4.445E-07	0.9995	
CD	1.27	1	1.27	0.3065	0.5845	
CE	0.5298	1	0.5298	0.1279	0.7235	
CF	0.0000	1	0.0000	7.073E-06	0.9979	
DE	155.28	1	155.28	37.47	< 0.0001	
DF	0.0001	1	0.0001	0.0000	0.9961	
EF	1.842E-06	1	1.842E-06	4.445E-07	0.9995	
A^2	0.6468	1	0.6468	0.1561	0.6960	
B^2	57.76	1	57.76	13.94	0.0009	
C^2	2.26	1	2.26	0.5465	0.4664	
D^2	2.22	1	2.22	0.5354	0.4709	
E^2	2349.88	1	2349.88	567.10	< 0.0001	
F^2	1.47	1	1.47	0.3559	0.5560	
Residual	107.74	26	4.14			
Lack of Fit	107.74	21	5.13			
Pure Error	0.0000	5	0.0000			
Cor Total	45038.25	53				

Adj R-Squared = 0.9976, Pred R-Squared = 0.9875, Adequate Precision = 67.77.

Table A-4
Analysis of variance for coolant temperature difference (ΔT_c).

Source	Sum of squares	df	Mean square	F-value	p-value	
Model	1251.35	27	46.35	424.66	< 0.0001	Significant
A-l	6.501E-07	1	6.501E-07	5.957E-06	0.9981	
B-h	0.2508	1	0.2508	2.30	0.1416	
C-t	0.2833	1	0.2833	2.60	0.1192	
D-s	2.501E-07	1	2.501E-07	2.292E-06	0.9988	
E-Q	1043.70	1	1043.70	9563.11	< 0.0001	
F-T_{in}	1.053E-07	1	1.053E-07	9.652E-07	0.9992	
AB	2.592E-07	1	2.592E-07	2.375E-06	0.9988	
AC	2.000E-08	1	2.000E-08	1.833E-07	0.9997	
AD	1.702E-07	1	1.702E-07	1.559E-06	0.9990	
AE	2.145E-07	1	2.145E-07	1.966E-06	0.9989	
AF	8.820E-08	1	8.820E-08	8.081E-07	0.9993	
BC	0.7501	1	0.7501	6.87	0.0144	
BD	9.800E-09	1	9.800E-09	8.979E-08	0.9998	
BE	0.4245	1	0.4245	3.89	0.0593	
BF	1.125E-10	1	1.125E-10	1.031E-09	1.0000	
CD	1.620E-08	1	1.620E-08	1.484E-07	0.9997	
CE	0.7501	1	0.7501	6.87	0.0144	
CF	5.062E-08	1	5.062E-08	4.639E-07	0.9995	
DE	3.781E-08	1	3.781E-08	3.465E-07	0.9995	
DF	6.845E-08	1	6.845E-08	6.272E-07	0.9994	
EF	1.125E-10	1	1.125E-10	1.031E-09	1.0000	
A^2	0.0132	1	0.0132	0.1207	0.7310	
B^2	0.0542	1	0.0542	0.4970	0.4871	
C^2	0.3380	1	0.3380	3.10	0.0902	
D^2	0.0133	1	0.0133	0.1216	0.7301	
E^2	160.77	1	160.77	1473.07	< 0.0001	
F^2	0.2144	1	0.2144	1.96	0.1728	
Residual	2.84	26	0.1091			
Lack of Fit	2.84	21	0.1351			
Pure Error	0.0000	5	0.0000			
Cor Total	1254.19	53				

Adj R-Squared = 0.9977, Pred R-Squared = 0.9881, Adequate Precision = 60.2987.

Appendix B. Final quadratic regression models

This appendix presents the final quadratic regression equations for the key performance metrics analyzed in this study:
Eq. B-1: Quadratic regression equation for average CPUs temperature (T_a).
Eq. B-2: Quadratic regression equation for maximum CPU temperature difference (ΔT_m).
Eq. B-3: Quadratic regression equation for coolant pumping power (P_P).
Eq. B-4: Quadratic regression equation for coolant temperature difference (ΔT_c).

Average CPUs temperature (T_a) $= 78.66 - 1.69 \times A - 3.2 \times B - 5.43 \times C - 8.74 \times D - 19.04 \times E + 10 \times F + 1.21 \times AB + 1.04 \times AC - 0.7983 \times AD + 1.24 \times AE + 0.0004 \times AF + 0.2586 \times BC - 6.33 \times BD + 1.3 \times BE - 0.0002 \times BF - 1.5 \times CD + 0.157 \times CE - 0.0006 \times CF + 5.23 \times DE - 0.0018 \times DF + 0.0002 \times EF - 0.7725 \times A^2 + 3 \times B^2 + 0.1352 \times C^2 + 4.97 \times D^2 + 4.99 \times E^2 - 4.99 \times F^2$ (B-1) Maximum CPUs temperature difference (ΔT_m) $= 10.20 + 2.51 \times A + 2.78 \times B - 0.2597 \times C + 0.2597 \times D - 6.18 \times E - 0.0017 \times F + 0.2281 \times AB - 0.2537 \times AC - 0.4237 \times AD - 0.8320 \times AE + 0.0011 \times AF + 0.3875 \times BC - 1.47 \times BD - 1.34 \times BE + 0.0003 \times BF - 0.3676 \times CD + 0.1487 \times CE + 0.0021 \times CF + 0.3812 \times DE + 0.0038 \times DF - 0.0003 \times EF - 0.2108 \times A^2 - 0.4326 \times B^2 + 0.1540 \times C^2 - 0.9956 \times D^2 + 3.15 \times E^2 - 0.0479 \times F^2$ (B-2).

Coolant pumping power (P_P) $= 30.16 + 3.16 \times A + 8.81 \times B - 0.0738 \times C - 3.55 \times D + 39.88 \times E - 0.0012 \times F + 0.5878 \times AB + 0.0402 \times AC + 0.2021 \times AD + 3.63 \times AE - 0.0004 \times AF - 0.4525 \times BC - 1.44 \times BD + 8.98 \times BE + 0.0005 \times BF - 0.3985 \times CD - 0.2573 \times CE + 0.0014 \times CF - 4.41 \times DE + 0.0036 \times DF - 0.0005 \times EF - 0.2508 \times A^2 - 2.37 \times B^2 + 0.4692 \times C^2 - 0.4644 \times D^2 + 15.11 \times E^2 - 0.3786 \times F^2$ (B-3).

Coolant temperature difference (ΔT_c) $= 7.13 + 0.0002 \times A + 0.1022 \times B + 0.1087 \times C - 0.0001 \times D - 6.59 \times E - 0.0001 \times F + 0.0002 \times AB + 0.0001 \times AC - 0.0001 \times AD - 0.0002 \times AE - 0.0001 \times AF - 0.3062 \times BC - 0.0000 \times BD - 0.1629 \times BE + 3.750\text{E-}06 \times BF + 0.0000 \times CD - 0.3062 \times CE + 0.0001 \times CF + 0.0001 \times DE + 0.0001 \times DF - 3.750\text{E-}06 \times EF + 0.0358 \times A^2 - 0.0726 \times B^2 - 0.1813 \times C^2 + 0.0359 \times D^2 + 3.95 \times E^2 + 0.1444 \times F^2$(B-4).

Data availability

Data will be made available on request.

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